

The Ledger Data Layer

Static, Reference, Market, Oracle, Smart-Contract Execution, and Listed-Instrument
Data: A Specification

Companion Document to the Ledger Framework Specification

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Abstract

The Ledger framework [1] defines portfolio state, conservation, and time travel in terms of *wallets*, *units*, *moves*, and *transactions*; the Valuation companion [2] specifies the P_t pricing function consumed by the PnL computation. Both presuppose a substrate of inputs — product terms, instrument masters, party identifiers, calendars, market observations, oracle attestations, calibrated objects, and the move stream itself. This document specifies that substrate as the *data layer*: a closed catalogue of nineteen leaves $L_1, \dots, L_{19}$ partitioned into six mutation-discipline classes, equipped with bitemporal indexing on every observation and definitional axis, governed by fifteen cross-cutting consistency laws $\Lambda_1, \dots, \Lambda_{15}$, seventeen determinism boundaries $B_1, \dots, B_{17}$, and six compositional theorems Th-1 through Th-6. The leaf taxonomy is built to a minimalist criterion: every concept that does not earn a place as an irreducible data category is folded into a parent leaf as a field, or relegated to a versioned sidecar. Two bitemporal axes (t_{obs} , t_{known}) are mandatory on every leaf in the *Definitions* and *Observations* classes; restatement is structural, not destructive. A *versioning algebra* pins seven mutation axes (component, schema, contract, model, refdata, regulator rule-set, canonicalisation) and binds each axis explicitly to the laws it discharges. An *operational floor* — per-leaf reconciliation pair, BreakRegister finite-state machine, retention matrix, service-level matrix, IPV/FRTB AVA fields, CSDR penalty schema — closes the gap between formal specification and production hygiene. A *realism budget* of eight unconditional guarantees and thirteen named conditional assumptions, each with detection signal, compensating action, blast radius, and a named owner, makes the trust frontier auditable. The Common Domain Model (CDM) is treated as a load-bearing vocabulary: a re-fetched cross-walk classifies every leaf as Direct, Partial, or Missing against FINOS CDM 6.0.0, and approximately fifteen pull-request-sized extensions are enumerated. The document is a specification, not a tutorial: definitions precede theorems, every claim is proven or its proof is named, and notation is fixed once and for all.

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1 Introduction

1.1 The Problem

The Ledger framework [1] reduces post-trade activity to a single primitive: the atomic move of a quantity between wallets. The Valuation companion [2] extends this with a pricing function $P_t(u)$ produced by a calibrated DAG. Both treat their inputs — product definitions, market observations, oracle attestations, party identifiers — as external. That choice was deliberate: a closed system is well-defined only when its frontier is named.

This document names the frontier. It specifies the data substrate that the Ledger and the Valuation stack consume, partitions it into a catalogue of irreducible leaves, fixes the bitemporal model under which observations may be restated without destruction, equips every leaf with a workflow shape and a CDM cross-walk, and binds the whole to fifteen cross-cutting consistency laws and seventeen determinism boundaries.

The data layer is not an integration concern that can be deferred. Without a precise data layer the conservation theorem $\sum_w w(u) = 0$ is a syntactic claim rather than a property of a running system; the path-independence of PnL is a property of an idealised price vector rather than a property of prices that arrive late, are restated, and are subject to vendor opacity; and the time-travel claim collapses the moment a vendor publishes a correction to a price already used to value a position. The data layer makes the abstract guarantees executable.

1.2 Scope

The data layer comprises **static data** (product terms, calendars, conventions), **reference data** (instrument masters, party identifiers, legal agreements, clock authorities), **market observations** (raw and calibrated), **oracle attestations** (lifecycle, external confirmations), **smart-contract execution data** (the canonical move stream, valuation records, obligations, regulatory submissions, break records), and **listed-instrument detail** (which is *not* a separate top-level category; see Section 3 and the veto register Section 6).

In scope:

1. The catalogue of leaves $L_1, \dots, L_{19}$ and their per-class mutation discipline.
2. The bitemporal model $(t_{\text{obs}}, t_{\text{known}})$ and the versioning algebra.
3. Per-leaf type design (closed sums, smart constructors, witness types) and CDM cross-walk.
4. Cross-leaf consistency laws $\Lambda_1, \dots, \Lambda_{15}$ and determinism boundaries $B_1, \dots, B_{17}$.
5. Compositional theorems Th-1 through Th-6.
6. The operational floor: reconciliation, breaks, retention, SLAs, IPV/FRTB AVA, CSDR penalty.
7. The property-based testing program and Goodhart hardening.

Out of scope: implementation language; storage technology choice; vendor selection; specific pricing methodology (treated in [2]); legal-entity onboarding workflows; KYC/AML controls beyond the data fields they require.

1.3 The Ledger Boundary

The data layer sits inside the closed-system boundary of the Ledger [1, §Scope and Limitations]. Authorities outside the boundary — GLEIF, ISO, SWIFT, vendors, regulators, calculation agents, central counterparties, custodians, trade repositories — are modelled as *attestors* whose statements enter the Ledger only through bitemporal observation leaves with explicit attestation envelopes. The data layer makes the boundary mechanical: every datum carries an attestor identifier, a signature, a known-time, and a snapshot reference, and the closed-system invariant of [1] reduces to the absence of writes that bypass these envelopes.

1.4 The Role of CDM

The Common Domain Model (CDM) plays the same role in the data layer that it plays in [1, §CDM Integration]: a canonical machine-executable vocabulary. The data layer is more demanding of CDM than the Ledger, because the data layer must classify every leaf as *Direct* (CDM is the wire and storage representation), *Partial* (CDM is the wire representation; storage extends), or *Missing* (Ledger-native; CDM extension proposed). Section 11 re-fetches the CDM 6.0.0 surface, classifies every leaf, and enumerates the approximately fifteen pull-request-sized extensions required.

1.5 Document Roadmap

Section 2 fixes notation and conventions. Section 3 states the master taxonomy of nineteen leaves and partitions them into six mutation-discipline classes. Section 4 gives per-leaf integrated specifications. Section 5 defines the bitemporal model and the versioning algebra. Section 6 states the engineering principles $P_1, \dots, P_{10}$ and the anti-over-engineering vetoes $V_1, \dots, V_{14}$ with truth conditions and CI mechanisms. Section 7 formulates the fifteen cross-cutting consistency laws. Section 8 catalogues the seventeen determinism boundaries. Section 9 states and proves the six compositional theorems. Section 10 specifies the realism budget. Section 11 performs the CDM gap analysis. Section 12 defines the operational floor: reconciliation matrix, break finite-state machine, retention matrix, SLA matrix, IPV/FRTB AVA fields, CSDR penalty. Section 13 declares the property-based testing program, the test pyramid, and the mutation kill-rate floors. Section 14 hardens against five Goodhart traps. Section 15 records the resolved structural choices. Section 16 contains the architecture decision register. Section 17 states the lines-of-code budget. Section 18 prescribes the verification approach. Appendix 19 is the glossary.

2 Notation, Conventions, and Glossary Cross-Reference

2.1 Naming-Scheme Disambiguation

Earlier drafts overloaded the prefix $L\#$ onto three distinct families. This document disambiguates aggressively. The fifteen prefixes used throughout are:

Prefix	Meaning
L_n	Leaf in the master taxonomy ($n = 1, \dots, 19$)
Λ_n	Cross-cutting consistency law ($n = 1, \dots, 15$)
Φ_n	Per-leaf invariant family (16 families $\times \{T, W, C\}$)
$\text{Th-}n$	Compositional theorem ($n = 1, \dots, 6$, with Th-4 split into 4a, 4b)
$\text{Lm-}n$	Lemma in the dependency directed acyclic graph
A_n	Axiom in the dependency directed acyclic graph
B_n	Determinism boundary ($n = 1, \dots, 17$)
$C\text{-}A_n$	Conditional realism-budget assumption ($n = 1, \dots, 13$)
U_n	Unconditional realism-budget guarantee ($n = 1, \dots, 8$)
N_n	Minimum data-quality requirement
P_n	Engineering principle ($n = 1, \dots, 10$)
V_n	Anti-over-engineering veto ($n = 1, \dots, 14$)
$M\text{-}n$	Mutation operator ($n = 1, \dots, 15$)
GT_n	Goodhart trap ($n = 1, \dots, 5$)
C_n	StatesHome structural index ($n = 1, \dots, 12$, per [1])
$\mathcal{D}_n$	Mutation-discipline class ($n = 1, \dots, 6$; introduced in §3)
$\text{ADR-}n$	Architectural Decision Record ($n = 1, \dots, 12$)

The two indices T (themes) and D (disagreements) used in the proposal trail are not load-bearing in this document and are dropped. The classes $\mathcal{D}_n$ are deliberately distinguished by font from the StatesHome C_n structural indices to avoid the collision that survived earlier drafts.

2.2 Bitemporal Axes

Every leaf in classes $\mathcal{D}_1$ (Definitions) and $\mathcal{D}_4$ (Observations) carries two timestamps:

$t_{\text{obs}} \in \text{ObsTime}$

The wall-clock instant the underlying real-world event occurred or is asserted to have occurred. Resolution: nanoseconds. Time zone: UTC by canonical convention. Tie-break: lexicographic on $(t_{\text{obs}}^{\text{ns}}, \text{attestor LEI, vendor message identifier})$.

$t_{\text{known}} \in \text{KnownTime}$

The wall-clock instant the Ledger first admitted the datum. Resolution: nanoseconds. UTC. Tie-break: ledger gateway commit order.

The constraints

$$t_{\text{obs}} \leq t_{\text{known}}, \quad t_{\text{known}} \leq \text{now}()$$

are enforced strictly. The first forbids future observations being admitted retroactively; the second forbids future-dated knowledge.

The two axes are not interchangeable. A query at a single time argument is ambiguous; the data layer exposes two orthogonal modes.

Definition 2.1 (Bitemporal Query Modes). *For a leaf $L \in C_1 \cup C_4$ carrying records of payload type T :*

1. Mode (a) — as-of: *as_of*(t_k) returns the snapshot of L as it was known at t_k . The result reflects what the system knew at t_k , not what was true.
2. Mode (b) — with-corrections-through: *with_corrections_through*(t_o, t'_k) returns the best estimate, as of knowledge time t'_k , of the truth at observation time t_o . The result reflects current best estimate, not historical knowledge.

Neither mode is derivable from the other; both are first-class.

2.3 StatesHome Indices

The twelve StatesHome structural indices $C_1, \dots, C_{12}$ are inherited from the Ledger [1, §StatesHome addendum]. The indices most frequently invoked here are: C_1 monotone-carrier; C_2 per-class structural $\sum = 0$; C_3 atomic StateDelta; C_4 capability-scoped reads; C_5 UnitStatus mutability; C_6 ProductTerms registration totality; C_7 ProductTerms append-only; C_8 two-track amendment via *is_fungibility_preserving*; C_{11} writer-cap per field; C_{12} overlay-keying schema.

2.4 Minimum Data-Quality Requirements (N_n)

The data-quality bar promised by the prefix table is enumerated below. Every workflow that ingests, transforms, or persists a leaf must satisfy these requirements. Detail is in the NAZAROV Phase 2 v2 contribution; this table is the normative index.

Requirement	Statement (summary)
N_1 Provenance	Every datum carries authority, channel, source-message-id, gateway, gateway-id
N_2 Attestation	Every observation carries a verifiable cryptographic envelope at the bound
N_3 Ingress validation	Total over a documented input domain; failure is a typed event, not a silent
N_4 Idempotency on replay	Total within a named scope; re-presenting a token returns the memoised c
N_5 Dispute resolution	Replay primitive (a)/(b) over the bitemporal axes; correction is a new eve
N_6 Point-in-time	Bitemporal mandatory on Definitions and Observations; both query mode
N_7 Failure mode named	Absence and contradiction are explicit events; silent fallback forbidden.
N_8 Multi-source aggregation	Per-leaf rule: aggregation function, disagreement threshold, quorum, fallba
$N_{8.2}$ Single-source escape	Permitted only with a registered authority assumption referenced in the tr
N_9 Mutable history forbidden	Restatements and corrections are new rows; archival is a query, not a mod
N_{10} Determinism	Snapshots are content-addressed; replay reconstructs the actually-traverse
N_{11} Mapping is part of the oracle	Schema mapping is total, version-pinned, bit-identical on replay.
N_{12} Trust assumptions first-class	Every untyped trust assumption is in the trust registry with detection sig

The handler-delay bound δ_h used in Theorem 9.3 is a separate symbol (not an N_n entry) and denotes a bounded propagation latency of the Temporal handler graph, not a data-quality requirement.

2.5 Glossary Cross-Reference

Terms used throughout this document that are not defined inline are defined in Appendix 19. The principal terms are: *Pricing DAG* (cf. [2, §6]); *mandate-as-unit*, *ProductTerms*, *UnitStatus*, *PositionState* (cf. [1, StatesHome]); *QIS* (Quantitative Investment Strategy unit type); *KIKO* (Knock-In/Knock-Out option lifecycle); *FpML*, *CDM* (FpML is XML schema; CDM is the FI-NOS Common Domain Model expressed in the Rosetta DSL); *StatesHome 3-map* (the maps *ProductTerms*, *UnitStatus*, *PositionState*[(w, u)]); *now()* (the boundary B_1 wall-clock read); Θ_{AF} (model-versioned no-arbitrage admissible parameter set, per Theorem 9.6); *JCS canonicalisation* (JSON canonicalisation per RFC 8785).

2.6 Typographical Conventions

Code identifiers and type names are typeset in **monospace**. Closed sums are written with the constructors enumerated; an ellipsis “...” inside a sum is forbidden in this document and a violation in any normative artefact. Arrows in equations bind to the right; quantifiers are explicit; the sentence “it is obvious that” is forbidden.

3 The Nineteen-Leaf Master Taxonomy

3.1 Construction Criterion

The taxonomy is constructed under one criterion: a concept appears as a top-level leaf if and only if it is an irreducible data category, that is:

1. It carries economic substance (it is not merely a wrapper or envelope discipline),
2. It admits a single mutation discipline (append-only versioned, mutable per-unit, monotone-carrier per pair, append-only bitemporal, append-only hash-chained), and
3. Its absence would force at least one cross-cutting law Λ_n to lose its structural anchor.

A concept that is a wrapper discipline (an attestation envelope, an idempotency token), a constants module (identity keys), a property of writes (a capability), or an aggregation of other leaves (a snapshot view) does *not* earn a top-level slot. It is folded into a parent leaf as a field, or relegated to a versioned sidecar. Two such sidecars are named throughout the document and are explicitly *not leaves*: *IdempotencyTokenSidecar* (the closed-sum cross-cutting field carried with every inbound payload) and *VersionPinSidecar* (the seven-axis versioning algebra of Section 5.2). Where earlier drafts wrote “ L_{20} ” or “ L_{21} ” they referred to these sidecars; the present document does not use those numerals.

Earlier drafts proposed twenty-four leaves. Five of those were folded; one was deleted; one was reclassified as a named view; one was decomposed. The collapse is documented in Section 3.4. Three load-bearing additions L_{17} , L_{18} , L_{19} are documented in Section 3.5. The final count is nineteen.

3.2 Six Mutation-Discipline Classes

The nineteen leaves partition into six classes, each carrying a uniform mutation discipline:

$\mathcal{D}_1$ Definitions

Append-only versioned. New versions are added; old versions are never mutated or deleted.
Members: $L_1, L_2, L_3, L_4, L_7^P, L_8, L_{19}$.

$\mathcal{D}_2$ Shared Status

Mutable per unit, single-writer-per-field. Members: L_5 .

$\mathcal{D}_3$ Per-Position State

Monotone-carrier per (w, u) pair. Members: L_6 .

$\mathcal{D}_4$ Observations

Bitemporal append-only. Members: $L_9, L_{10}, L_{11}, L_{12}$.

$\mathcal{D}_5$ Effects

Append-only hash-chained (or append-only with reconciliation cadence). Members: $L_{13}, L_{14}, L_{15}, L_{17}, L_{18}$.

$\mathcal{D}_6$ Provenance

Cross-cutting authoritative tables not subsumed elsewhere. Members: L_{16} .

3.3 The Catalogue

Leaf	Name	Class	One-line characterisation
L_1	ProductTerms	$\mathcal{D}_1$	StatesHome map 1; CDM
L_2	InstrumentMaster	$\mathcal{D}_1$	NonTransferableProduct-aligned Multi-vendor reconciled instrument identity

Leaf	Name	Class	One-line characterisation
L_3	PartyLEI	$\mathcal{D}_1$	GLEIF-anchored identifier with BIC/MIC indexes
L_4	CalendarConvention	$\mathcal{D}_1$	Calendars and day-counts; mode-1 amendment pin
L_5	UnitStatus	$\mathcal{D}_2$	StatesHome map 2; mutable; single-writer-per-field
L_6	PositionState	$\mathcal{D}_3$	StatesHome map 3; six-coordinate vector
L_7^P	PolicyConfiguration	$\mathcal{D}_1$	Partitioned $L_7^{Pa}/L_7^{Pb}/L_7^{Pc}$
L_8	LegalAgreement	$\mathcal{D}_1$	ISDA Master, CSA, GMSLA, mandate
L_9	RawMarketObservation	$\mathcal{D}_4$	Bitemporal stream with N_8 aggregation gate
L_{10}	LifecycleOracle	$\mathcal{D}_4$	18-constructor closed sum of lifecycle events
L_{11}	ExternalConfirmation	$\mathcal{D}_4$	ISO 20022 / FpML inbound message stream
L_{12}	CalibratedMarketObject	$\mathcal{D}_4$	Kalman posterior with arbitrage certificate
L_{13}	MoveStream	$\mathcal{D}_5$	Canonical hash-chained transaction sequence
L_{14}	ValuationRecord	$\mathcal{D}_5$	Per- (u, t, model) valuation with IPV/AVA fields
L_{15}	Obligation	$\mathcal{D}_5$	Pending $\rightarrow$ Discharged Compensated Defaulted FSM
L_{16}	ReferenceMaster	$\mathcal{D}_6$	Cross-cutting authoritative tables
L_{17}	RegulatorySubmission	$\mathcal{D}_5$	Append-only per submission; bitemporal restatement chain
L_{18}	BreakRegister	$\mathcal{D}_5$	Mutable break record with FSM
L_{19}	ClockAuthority	$\mathcal{D}_1$	Bitemporal time-source authority

3.4 The Collapse

The leaves removed from earlier drafts and their disposition:

Earlier leaf	Disposition	Rationale
AttestationEnvelope	Folded as field	Wrapper discipline, not data category
IdentityKeys	Folded as field	Constants module
IdempotencyToken	Folded as field	Cross-cutting field with closed-sum kind
HashChainAnchor	Folded into L_{13}	Property of L_{13} , not separate
OrchestrationState	Deleted	V_{11} violation; not economic data
Snapshot	Reclassified	Aggregation of $L_9 + L_{12}$; named view, not stored leaf
Capability	Decomposed	Closed-alphabet field_tag enum + per-leaf writer_cap phantom

3.5 The Three Load-Bearing Additions

*L*₁₇ **RegulatorySubmission.** Append-only per `submission_id`. Carries (`regulator`, `rule_set`, `rule_set_version`). Justified by ADR-2: CFTC, EMIR Refit, SFTR, SLATE, MiFIR RTS 22, and FRTB Pillar 3 reporting are not derivable from the move stream alone; they require a versioned, regulator-typed, acknowledged artefact whose lineage is auditable.

*L*₁₈ **BreakRegister.** Mutable per `break_id` with the finite-state machine of Section 12.2. Justified by ADR-3: SOX §404, BCBS 239 §3, and DORA Article 8 require a first-class break-management artefact; recording breaks as ordinary observations would lose the FSM and four-eyes discipline.

*L*₁₉ **ClockAuthority.** Bitemporal per `authority_id`. Carries (`source_kind` $\in$ {`NTP`, `PTP`, `GNSS`, `atomic`}, `leap`). Every t_{obs} and t_{known} in *L*₉, *L*₁₀, *L*₁₁, *L*₁₃ references it. Justified by ADR-4: the time-translation invariance carrier in [1, §Replay determinism] requires an explicit authoritative source; replay determinism is otherwise dependent on operating-system clock state.

3.6 Architecture Decision Register Index

The twelve architecture decisions referenced throughout this section, and developed elsewhere in the document, are catalogued in Section 16. The numbering is fixed: ADR-1 through ADR-12.

4 Per-Leaf Integrated Specifications

For every leaf this section gives a uniform set of fields. The fields are: **(a)** the carrier and well-formedness predicate (a definition); **(b)** the type design (a closed sum and smart constructor where appropriate); **(c)** the workflow shape (idempotency key, replay-determinism class); **(d)** the CDM cross-walk (Direct, Partial, or Missing) with concrete CDM type names; **(e)** the principal invariants Φ ; **(f)** the cross-cutting laws Λ in which the leaf appears; **(g)** the reconciliation pair (external authoritative source, cadence, tolerance, break workflow, owner); **(h)** the realism-budget anchor.

Two paragraphs of prose per leaf are provided. This document is the authoritative consolidated record of the structured fields; further detail (Phase 2 specialist contributions and the round-by-round adversarial-review trail) is preserved in the project work folder for archival reference but is not normative.

4.1 L_1 ProductTerms

Definition 4.1 (L_1 Carrier and well-formedness). L_1 is a map keyed by `unit_id`, valued in `NonEmpty<TermsVersion>`. The well-formedness predicate requires: (i) append-only versioning (the version sequence is a totally ordered prefix of the unit's history); (ii) t_{known} monotone non-decreasing along the version sequence; (iii) the function `is_fungibility_preserving : TermsVersion → TermsVersion → Bool` is total.

Type design. Closed sum:

```
type TermsVersion =
  | ImmutableTerms { fields, signed_by, hash }
  | FungibilityPreservingAmendment { effective_at, knowledge_at,
    fields, preservation_witness, signed_by, hash }
  | FungibilityBreakingAmendment { effective_at, knowledge_at,
    fields, break_reason: BreakReason, signed_by, hash }

type BreakReason =
  | UnderlierIdentityChange | StrikeTermsChanged
  | MaturityChanged | CashflowFunctionChanged
  | RegulatoryRecategorisation
  | CounterpartyAssignmentBreakingFungibility
```

The predicate-as-field of earlier drafts is gone: the constructor `is` the fungibility status. **Workflow shape.** `RegisterUnitWorkflow` (idempotency key `unit_id = hash_jcs(canonical terms blob)`); replay-determinism class: content-addressed. **CDM cross-walk.** Direct on `NonTransferableProduct` (file `product-template-type.rosetta`); the qualification step uses CDM `Qualify_*` functions. **Invariants.** Φ_1^T totality of `is_fungibility_preserving`; Φ_1^W append-only versioning; Φ_1^C smart-contract reference resolves. **Laws.** Anchors $\Lambda_2, \Lambda_4, \Lambda_9$. **Reconciliation pair.** (Counterparty CDM `BusinessEvent`, per-trade, bit-identical, `wf-trade-affirm-break`, front-office trade support). **Realism anchor.** U_1 append-only mutation; C-A₅ CDM/ISO/FpML schema stability.

4.2 L_2 InstrumentMaster

Definition 4.2 (L_2 Carrier and well-formedness). L_2 is a bitemporal map keyed by (`authority, instrument_id, version`). Well-formedness: (i) $t_{\text{obs}} \leq t_{\text{known}}$; (ii) the multi-vendor reconciliation gate N_8 has fired; (iii) the aggregation outcome is recorded explicitly.

Type design. Bitemporal record carrying `authority, instrument_id, version, terms_ref, aggregation_outcome ∈ {multi_source_consensus, unique_authority, quarantined}`. **Work-**

flow shape. Daily T+0 cross-vendor reconciliation activity. **CDM cross-walk.** Partial: CDM Security extends InstrumentBase is direct on identity; missing on operational fields (boardLotSize, votingRights, dividendPolicyRef, corporateActionFeedRef); Section 11 enumerates the extension. **Invariants.** Φ_2^T identity uniqueness; Φ_2^C aggregation outcome present. **Laws.** Anchors Λ_4 (bitemporal coherence). **Reconciliation pair.** (Cross-vendor IM, daily, 0 mismatch tolerance, wf-refdata-break, refdata operations). **Realism anchor.** C-A₃ vendor honesty; C-A₇ authority registry currency.

4.3 L_3 PartyLEI

Definition 4.3 (L_3 Carrier and well-formedness). L_3 is a bitemporal map keyed by LEI with secondary indexes BIC and MIC. Well-formedness: (i) the GLEIF signature verifies; (ii) *status* $\in \{\text{Issued, Lapsed, Retired, Annulled, Duplicate}\}$ (closed sum).

Type design. Closed-sum status; LEI conforms to ISO 17442. **Workflow shape.** Daily refresh against GLEIF Concatenated Data File; status-change signal triggers downstream FSM transitions. **CDM cross-walk.** Direct on CDM Party (file base-staticdata-party-type.rosetta). **Invariants.** Φ_3^T closed-sum status; Φ_3^W signature verifies. **Laws.** Anchors Λ_4, Λ_{14} . **Reconciliation pair.** (GLEIF CDF, daily, status match, wf-lei-break, identity-and-trust ops). **Realism anchor.** C-A₇.

4.4 L_4 CalendarConvention

Definition 4.4 (L_4 Carrier and well-formedness). L_4 is a bitemporal map keyed by (*BusinessCenter*, *year-range*). Well-formedness: (i) mode-1 amendment pin: a calendar republication produces a new bitemporal record but never invalidates a pinned schedule on a unit registered before the republication.

Type design. Calendar payload + closed-sum DayCountFractionEnum, BusinessDayConventionEnum. **Workflow shape.** Calendar restatements flow through RestatementWatchWorkflow (Section 12.4); subscribers see a signal but the mode-1 pin guarantees that already-fired timers are untouched. **CDM cross-walk.** Direct on CDM BusinessCenters, BusinessDayConventions, DayCountFractionEnum. **Invariants.** Φ_4^T mode-1 pin; Φ_4^W bitemporal coherence. **Laws.** Anchors Λ_4 . **Reconciliation pair.** (Multi-vendor, daily, 0, wf-calendar-break, refdata operations). **Realism anchor.** C-A₇.

4.5 L_5 UnitStatus

Definition 4.5 (L_5 Carrier and well-formedness). L_5 is a mutable map keyed by *unit_id*. Each field carries a *writer_cap* phantom; well-formedness requires single-writer-per-field per StatesHome C_{11} .

Type design. Per-field closed-sum state. The triggered-barrier flag of earlier drafts is replaced by BarrierState = NotBreached | Breached { observation_event_ref, breach_t_obs, attestor }. **Workflow shape.** Per-state-change activity invoked by lifecycle workflows; ADR-1 carve-out of veto V_{13} . **CDM cross-walk.** Partial: CDM has lifecycle event types, not a directly-aligned mutable status map; mapping is via TradeState.state. **Invariants.** Φ_5^T writer-cap respected; Φ_5^W closed-sum field values; Φ_5^C cache coherence with L_{13} (Theorem 9.5). **Laws.** Anchors Λ_4, Λ_{14} . **Reconciliation pair.** (Counterparty FpML / sese.025, per state change, bit-identical, wf-state-break, lifecycle operations). **Realism anchor.** U_7 single-writer-per-field.

4.6 L_6 PositionState

Definition 4.6 (L_6 Carrier and well-formedness). L_6 is a map keyed by $(w, u) \in \mathcal{W} \times \mathcal{U}$. For lendable securities the value is the six-coordinate vector (own, onloan, borr, coll_post, coll_recv, coll_rehyp) $\in \mathbb{R}^6$, subject to the closure law $\text{coll_rehyp} \leq \text{coll_recv}$ and $\text{borr} \geq 0$ (cf. [1, §Generalised Position Model]). For non-lendable units the vector degenerates to a scalar.

Type design. Smart constructor `mk_gpm6` rejects $\text{coll_rehyp} > \text{coll_recv}$ and $\text{borr} < 0$; `apply_move` enforces the Single-Coordinate Move Principle at the type level. **Workflow shape.** Updates flow through the executor activity invoked by lifecycle workflows; bitemporal not required (L_6 is a stored cache). **CDM cross-walk.** Partial: CDM has `Position`; the six-coordinate decomposition is Ledger-native. **Invariants.** Φ_6^T closure law; Φ_6^W Single-Coordinate Move Principle; $\Phi_6^C \text{ avail} = \text{own} - \text{onloan} + \text{borr}$ (definitional identity). **Laws.** Anchors $\Lambda_5, \Lambda_6, \Lambda_7, \Lambda_{15}$. **Reconciliation pair.** (CCP daily statement, custodian, triparty agent; daily T+1; per-regime tolerance; `wf-position-break`; middle-office reconciliation). **Realism anchor.** U_7 . **Sub-leaf register.** L_6 carries fourteen SBL sub-leaves $L_{6.1}, \dots, L_{6.14}$ (locate, cascade-recall, rehyp-cap, regulatory cursor, fee-quote, rebate, manufactured-payment, RQV, MMF NAV, locate-confirmation, buy-in, triparty agreement, agent-lender disclosure, SBL loan unit state); see [1, §SBL] for the full register.

4.7 L_7^P PolicyConfiguration

Definition 4.7 (L_7^P Partition). L_7^P partitions into three sidecars, each capped at thirty fields per partition ($ADR\text{-}12$, V_9 enforcement):

- L_7^{Pa} : firm currency, decimal precision, rounding mode.
- L_7^{Pb} : tolerance thresholds ($PnL\text{-}explain$, reconciliation, $FRTB\text{ PLA}$).
- L_7^{Pc} : capability schema, version pins, retention horizons.

Bootstrap. The `L7Pc@genesis` entry sits in the genesis hash anchor; all subsequent updates are L_{13} transactions referencing prior L_7^{Pc} by `tx_id`. **Workflow shape.** Policy changes are governance-gated; deployment activity. **CDM cross-walk.** Missing (Ledger-internal). **Invariants.** Φ_7^W field-cap respected; Φ_7^C genesis bootstrap intact. **Laws.** Anchors Λ_8, Λ_{14} . **Reconciliation pair.** Internal governance audit; per-change; bit-identical; `wf-policy-break`; architecture review board. **Realism anchor.** U_6 schema-pinned validation.

4.8 L_8 LegalAgreement

Definition 4.8 (L_8 Carrier and well-formedness). L_8 is a map keyed by `agreement_id`. Well-formedness: (i) bilateral signature confirmation present; (ii) `document_hash` matches the signed PDF anchor.

Type design. Closed sum on `collateral_regime` (`TitleTransfer`, `SecurityInterest`, `UsRule15c3_3`); `governing_law` closed sum. **Workflow shape.** Activity-only registration (no orchestration); change-control via four-eyes. **CDM cross-walk.** Direct on CDM `LegalAgreement` with `LegalAgreementTypeEn`. **Invariants.** Φ_8^T regime closed sum; Φ_8^W signature verifies. **Laws.** Anchors Λ_3, Λ_{14} . **Reconciliation pair.** (Counterparty signed PDF, per execution, bit-identical, `wf-agreement-break`, legal). **Realism anchor.** C-A₈ closed-system boundary integrity.

4.9 L_9 RawMarketObservation

Definition 4.9 (L_9 Carrier and well-formedness). L_9 is an append-only bitemporal stream keyed by $(\text{topic}, t_{\text{obs}}, \text{source})$. Well-formedness: (i) the attestation envelope verifies; (ii) $t_{\text{obs}} \leq t_{\text{known}}$;

(iii) *aggregation_outcome* $\in \{\text{multi_source_consensus}, \text{unique_authority}, \text{quarantined}\}$ is recorded; (iv) admission into a snapshot consumed by L_{12} requires either *multi_source_consensus* or an explicit *single_source_authority_assumption_ref* into the trust registry.

Type design. Bitemporal record with closed-sum aggregation outcome. **Workflow shape.** Out-of-Temporal feed handlers write into the snapshot store; workflows consume via `capture_snapshot`. **CDM cross-walk.** Partial on CDM `Observation` (file `observable-event-type.rosetta`); CDM observations are scoped to lifecycle, not generic streaming ingest. **Invariants.** Φ_9^T envelope verifies; Φ_9^W aggregation outcome closed sum; Φ_9^C N8 gate before snapshot inclusion. **Laws.** Anchors Λ_1, Λ_4 . **Reconciliation pair.** (Multi-vendor IPV at fair-value hierarchy, daily/weekly per asset class, FRTB AVA tolerance, `wf-ipv-break`, IPV team). **Realism anchor.** C-A₃ vendor honesty; C-A₁ cryptographic primitive soundness.

4.10 L_{10} LifecycleOracle

Definition 4.10 (L_{10} Carrier and well-formedness). L_{10} is an append-only bitemporal stream of authority-signed lifecycle events. Well-formedness: (i) authority signature verifies; (ii) the constructor lies in the eighteen-element closed sum below; (iii) the idempotency key is content-addressed by *business_event_id*.

Type design. The closed sum:

```
type LifecycleEvent =
  | CorporateAction | Barrier | Fixing | Exercise
  | Default | Locate | RegulatoryThreshold | ForceMajeure
  | ManufacturedPayment | ResetObservation
  | Novation | ClearingNovation | BuyIn | Recall
  | RehypEvent | ConfirmationReceived
  | AgentDisclosure | LossEvent
```

Eighteen constructors; no ellipsis; the boundary parser rejects unknown tags with `UnknownTag { tag, known = [...] }`. **Workflow shape.** Per-event lifecycle workflow signalled by oracle ingress. **CDM cross-walk.** Partial on `BusinessEvent`, `Reset`, `ObservationEvent.corporateAction`; SBL events (`Locate`, `Recall`, `RehypEvent`, `AgentDisclosure`) Missing and tracked under the ISLA CDM working group. **Invariants.** Φ_{10}^T closed sum; Φ_{10}^W authority signature; Φ_{10}^C idempotency-key dedup. **Laws.** Anchors $\Lambda_3, \Lambda_4, \Lambda_{13}$. **Reconciliation pair.** (Calculation agent / index admin, per event, authoritative match, `wf-oracle-break`, lifecycle operations). **Realism anchor.** C-A₃.

4.11 L_{11} ExternalConfirmation

Definition 4.11 (L_{11} Carrier and well-formedness). L_{11} is an append-only stream keyed by (*transaction_id_ref*, *external_message_id*). Well-formedness: (i) wire signature verifies; (ii) *transaction_id_ref* resolves to an L_{13} entry.

Type design. ISO 20022 / FpML envelope payload; closed-sum *message_kind*. **Workflow shape.** Settlement orchestration consumes via signal; T+0 settlement deadline propagates as a workflow timer. **CDM cross-walk.** Partial on inbound mappings (`sese.025`, `sese.023`, `camt.053`, `camt.054`); outbound mapping is the regulatory-submission concern of L_{17} . **Invariants.** Φ_{11}^T wire signature; Φ_{11}^W *transaction_id_ref* resolves. **Laws.** Anchors Λ_3 . **Reconciliation pair.** (Inbound ISO 20022, per message, validation only, `wf-confirm-break`, settlement operations). **Realism anchor.** C-A₄ settlement-layer SSI freshness.

4.12 L_{12} CalibratedMarketObject

Definition 4.12 (L_{12} Carrier and well-formedness). L_{12} is a versioned snapshot per $(target_object, model_id, model_version_at_cert_time)$. *Well-formedness:* (i) admission to consumers is gated on **certified** = **true**; (ii) the witness type **arbitrage_certificate** is present; (iii) **model_version_at_cert_time** is recorded explicitly.

Type design. GADT-keyed by model:

```
type CalibrationOutcome =
  | Rejected { reason: RejectionReason }
  | Accepted { posterior: KalmanPosterior,
              certificate: arbitrage_certificate }
```

The posterior is reachable *only* inside the **Accepted** branch (witness type discipline). **Workflow shape.** Kalman calibration workflow with **ContinueAsNew** at $K = 200$ steps or 30 days; pinned **KalmanContinuePayload** v1. **CDM cross-walk.** Missing on content; CDM has the container **TradeState.valuationHistory** (cardinality $0..*$). **Invariants.** Φ_{12}^T certificate witness; Φ_{12}^W snapshot id content-addressed; Φ_{12}^C posterior-only-on-Accepted. **Laws.** Anchors $\Lambda_2, \Lambda_{11}, \Lambda_{12}$. **Reconciliation pair.** (Cross-engine IPV; PnL-explain residual, per repricing cycle, L_7^{Pb} tolerance, wf-cal-break, quant validation). **Realism anchor.** C-A₆ calibration model soundness.

4.13 L_{13} MoveStream

Definition 4.13 (L_{13} Carrier and well-formedness). L_{13} is an append-only hash-chained sequence of **BalancedTransaction**. *Well-formedness:* (i) per-class structural $\sum = 0$ (**StatesHome** C_2); (ii) $tx_id = \text{hash_jcs}(\text{business_event_id}, \text{attempt_seq})$, with **run_id** explicitly absent (**ADR-7**); (iii) **prev_hash** matches the predecessor.

Type design. **Transaction** = **NonEmpty** $\langle$ **Move** $\rangle$ with the conservation refinement **BalancedTransaction**; only **BalancedTransaction** is admissible to executor commit. **Workflow shape.** Per-business-event workflow; **attempt_seq** carried across **ContinueAsNew**. **CDM cross-walk.** Partial on CDM **BusinessEvent** + **TradeState**; the snapshot-id binding, executor signature, and hash-chain pointer are Ledger-native (CDM downgrade per Section 11). **Invariants.** Φ_{13}^T balanced; Φ_{13}^W hash-chain matches; Φ_{13}^C **tx_id** excludes **run_id**. **Laws.** Anchors $\Lambda_5, \Lambda_6, \Lambda_7, \Lambda_8, \Lambda_{10}, \Lambda_{15}$. **Reconciliation pair.** (Hash-chain self-verify; cross-replica, continuous, 0, wf-chain-break, architecture review). **Realism anchor.** U_3, U_4, U_8 .

4.14 L_{14} ValuationRecord

Definition 4.14 (L_{14} Carrier and well-formedness). L_{14} is versioned per $(unit_id, t, model_id)$. *Well-formedness:* (i) **attestation_snap** resolves to an L_{19} snapshot; (ii) **quality** $\in$ **ValuationQuality** (closed sum); (iii) the FSM state lies in $\{\text{PRICING, PRICED, STALE, FAILED, ...}\}$ per [2, §2].

Type design. Augmented schema (closing the IPV/AVA gap):

```
ValuationRecord:
  fair_value_level    in {1, 2, 3}
  ipv_status          IpvStatus
  ipv_variance        Decimal
  ipv_source_id       SourceId
  prudent_valuation_adjustment_components: PVAComponents
  unobservable_inputs []UnobservableInput
  unobservable_input_sensitivity []SensitivityRow
```

where **PVAComponents** carries **market_price_uncertainty**, **close_out_cost**, **model_risk**, **concentrated_position**, **future_admin_costs**, **early_termination**, **operational_risk**. **Workflow shape.** Per-unit

valuation workflow per [2, §6]. **CDM cross-walk.** Missing on content; the container is CDM `TradeState.valuationHistory`. **Invariants.** Φ_{14}^T FSM state closed; Φ_{14}^C snapshot resolves. **Laws.** Anchors $\Lambda_2, \Lambda_8, \Lambda_{11}$. **Reconciliation pair.** (Cross-engine IPV; Level 1/2/3 reconcile, daily, FRTB PLA tolerance, wf-valuation-break, risk). **Realism anchor.** C-A₆.

4.15 L_{15} Obligation

Definition 4.15 (L_{15} Carrier and well-formedness). L_{15} is a map keyed by *obligation_id* with finite-state machine $PENDING \rightarrow DISCHARGED \mid COMPENSATED \mid DEFAULTED$. Well-formedness: *discharge_predicate* and *compensation_handler* both lie in named closed sums.

```
type DischargePredicateKind =
  | ByDeadline { deadline: Timestamp }
  | ByMatch     { matcher: MatchSpec }
  | ByAttestation { authority: LEI, attestation_kind: ... }
  | ByCondition { evaluator: ConditionRef }

type CompensationHandlerKind =
  | TrivialActivity { handler_ref }
  | NonTrivialChildWorkflow { workflow_ref, tower_depth: 1|2|3 }
```

Workflow shape. ObligationWorkflow with three-tier saga compensation tower (Section 12.2). **CDM cross-walk.** Missing (Ledger-internal). **Invariants.** Φ_{15}^T FSM closed; Φ_{15}^W predicate evaluates bitemporally; Φ_{15}^C κ -totality across EventClass $\times$ ObligationKind. **Laws.** Anchors Λ_{13} . **Reconciliation pair.** (CSA-call, AcadiaSoft, triResolve, TR ack, regulator ack; per event; per obligation kind tolerance; wf-obligation-break; collateral operations). **Realism anchor.** U_8 replay determinism via composition; bounded horizon $T_{\max}$.

4.16 L_{16} ReferenceMaster

Definition 4.16 (L_{16} Carrier). L_{16} is the reserved cross-cutting catch-all for authoritative tables not subsumed by L_2, L_3, L_4, L_8 : sanctions lists, withholding tables, CCP margin parameters, index compositions.

Type design. A discriminated union over the constituent table kinds: `ReferenceMaster = Sanctions | Withholding | CorporateActions | IndexComposition | CcpMargin`. Each variant is a bitemporal map keyed appropriately. **Workflow shape.** Per-kind ingestion activities; multi-vendor reconciliation gate N_8 where two or more sources publish; bitemporal restatement otherwise. **CDM cross-walk.** Sanctions *Missing*; Withholding *Missing*; CorporateActions *Partial* via `CorporateActionEvent`; IndexComposition *Partial* via `Index / EquityIndex` (constituent set thin); CcpMargin *Missing*. **Invariants.** Φ_{16}^T each variant has a documented authoritative source; Φ_{16}^W ingress is gated by N_2 attestation; Φ_{16}^C cross-walk preserved on replay. **Laws.** Anchors Λ_4 (bitemporal coherence) and Λ_9 (forgetful-functor composition). **Reconciliation pair.** (Per-kind authoritative source, daily, 0 tolerance, wf-refdata-break, refdata operations). **Realism anchor.** $U_1, U_2, C-A_3$ (per-kind vendor honesty).

4.17 L_{17} RegulatorySubmission

Definition 4.17 (L_{17} Carrier and well-formedness). L_{17} is append-only per *submission_id* with bitemporal restatement chain. Well-formedness: (i) *drd_rule_set_version* pinned per

the versioning algebra; (ii) payload is CDM-native (the projection μ_{P_3} of Theorem 9.7); (iii) *acknowledgement_status* closed sum.

Workflow shape. ResubmissionWorkflow on restatement; deadline propagation per Section 12.4. **CDM cross-walk.** Partial: CDM DRR (Digital Regulatory Reporting) covers Pillar 3, MiFIR, EMIR Refit, SFTR, CFTC Part 49; rule-set version is pinned via the *rule-set* axis of the VersionPinSidecar (Section 5.2). **Invariants.** Φ_{17}^T rule-set version pinned; Φ_{17}^W ack closed sum; Φ_{17}^C projection μ_{P_3} deterministic. **Laws.** Anchors Λ_8, Λ_9 . **Reconciliation pair.** (Trade Repository ack, per submission, ack match, *wf-regsub-break*, regulatory operations). **Realism anchor.** U_1, U_4, U_6 .

4.18 L_{18} BreakRegister

Definition 4.18 (L_{18} Carrier and well-formedness). L_{18} is mutable per *break_id*. Well-formedness: state lies in the FSM

$OPEN \rightarrow INVESTIGATING \rightarrow ASSIGNED \rightarrow AGED-1 \rightarrow AGED-3 \rightarrow AGED-5 \rightarrow ESCALATED \rightarrow AT-RISK \rightarrow MATERIAL$

CLOSED-WAIVED requires four-eyes approval.

Workflow shape. Per-break workflow with aging timers (T+1, T+3, T+5); escalation to governance committee at MATERIAL. **CDM cross-walk.** Missing (Ledger-internal). **Invariants.** Φ_{18}^T FSM closed; Φ_{18}^W four-eyes on CLOSED-WAIVED; Φ_{18}^C aging monotone. **Laws.** Anchors Λ_8, Λ_{14} . **Reconciliation pair.** Per-leaf reconciliation cadence (cf. Table 7). **Realism anchor.** U_1 .

4.19 L_{19} ClockAuthority

Definition 4.19 (L_{19} Carrier and well-formedness). L_{19} is bitemporal per *authority_id*. Well-formedness: (i) *source_kind* closed sum; (ii) *leap_second_policy_version* pinned; (iii) *signature* verifies; (iv) every t_{obs} and t_{known} in $L_9, L_{10}, L_{11}, L_{13}$ references it.

Workflow shape. Periodic refresh activity; signal on offset change. **CDM cross-walk.** Missing (Ledger-internal). **Invariants.** Φ_{19}^T source-kind closed sum; Φ_{19}^W signature verifies; Φ_{19}^C leap-second policy pinned. **Laws.** Anchors Λ_4, Λ_8 (time-translation invariance). **Reconciliation pair.** Per-source attestation; continuous; 0; internal break. **Realism anchor.** C-A₇.

5 Bitemporal Model and Versioning Algebra

5.1 The Bitemporal Type

Definition 5.1 ($\text{Bitemporal}\langle T \rangle$). *For a payload type T , the type $\text{Bitemporal}\langle T \rangle$ is the parametric record*

```

type Bitemporal<T> = {
  records: List<BitemporalRecord<T>>,
}

type BitemporalRecord<T> = {
  payload:      T,
  t_obs:        ObsTime,
  t_known:      KnownTime,
  attestor:     LEI,
  signature:    Signature,
  restate_link: Option<RecordId>,
}

```

together with the query API of Definition 2.1, the totality of which is verified by the well-formedness predicate *well_formed*.

Worked restatement example. A vendor publishes $\text{EUR/USD} = 1.0852$ at $t_{\text{obs}} = 2026-04-27\text{T}15:30:00.000\text{Z}$, $t_{\text{known}} = 2026-04-27\text{T}15:30:00.123\text{Z}$. On $2026-04-30\text{T}09:00:00\text{Z}$ the vendor restates to 1.0853 with the *same* t_{obs} . Two bitemporal records:

R_1 : payload = 1.0852, $t_{\text{obs}} = 2026-04-27\text{T}15:30:00.000\text{Z}$, $t_{\text{known}} = 2026-04-27\text{T}15:30:00.123\text{Z}$, $\text{restate_link} = \perp$
 R_2 : payload = 1.0853, $t_{\text{obs}} = 2026-04-27\text{T}15:30:00.000\text{Z}$, $t_{\text{known}} = 2026-04-30\text{T}09:00:00.000\text{Z}$, $\text{restate_link} = R_1$

The query `as_of(2026-04-29T00:00:00Z)` returns 1.0852: as of that knowledge time, R_2 did not exist. The query `with_corrections_through(2026-04-27T15:30:00.000Z, 2026-04-30T10:00:00Z)` returns 1.0853: this is the best estimate, as of the later knowledge time, of the truth at the original observation time.

Invariant 5.2 (Bitemporal forbidden patterns). *The following operations are forbidden on $\text{Bitemporal}\langle T \rangle$:*

1. Mutating R_1 in place.
 2. Deleting R_1 .
 3. Producing a merged record without setting *restate_link*.
 4. Treating `as_of( $t_{\text{known}}$ )` as a function of t_{obs} alone.
- A violation in any normative artefact is a V_4 failure (canonical-storage discipline).

$\text{Bitemporal}\langle T \rangle$ is mandatory on every leaf in classes $\mathcal{D}_1$ (Definitions) and $\mathcal{D}_4$ (Observations). A single-axis “as-of” representation elsewhere is a violation of principle P_4 .

5.2 The Versioning Algebra

A single “version” is insufficient: at least seven distinct mutation axes coexist, and conflating them is a frequent source of replay-determinism bugs. The data layer pins them separately.

Definition 5.3 (Versioning Algebra). *The seven-axis pin record:*

```

type VersionPin = {
  component_pin:  Map<ComponentName, GitSha>,
  schema_pin:     Map<SchemaId, SchemaVersion>,
  contract_pin:   Map<ContractId, ContractVersion>,
  model_pin:      Map<ModelId, ModelVersion>,
}

```

```

    refdata_pin:      Map<RefDataAuthority, RefDataVersion>,
    drr_rule_set_pin:  Map<RegulatorRuleSet, GitSha>,
    canonicalisation_pin: Map<CanonicalisationDomain,
                             RFC8785 | ProtobufCanonical | CBORProfile>,
  }

```

The mutation discipline per axis:

- `component_pin`: append-only on deploy; old SHAs retained for replay.
- `schema_pin`: append-only with versioned closed enum; per-pair migration table.
- `contract_pin`: append-only; `is_fungibility_preserving` decides amendment branch.
- `model_pin`: append-only; pinned at calibration time; `model_version_at_cert_time` carried in L_{12} .
- `refdata_pin`: bitemporal; restated rather than mutated.
- `drr_rule_set_pin`: append-only per regulator; rule-set version travels with L_{17} .
- `canonicalisation_pin`: append-only; canonicalisation version is its own axis.

Remark 5.4 (Axes required per law). *For each law Λ_n , the data layer specifies which axes must be unchanged for the law to hold under replay. Examples:*

Λ_8 (Replay Determinism): *`component_pin` $\wedge$ `schema_pin` $\wedge$ `canonicalisation_pin` unchanged.*

Λ_{12} (No-Arbitrage): *`model_pin` $\wedge$ `schema_pin` unchanged at the calibration time of every certified L_{12} .*

Λ_9 (Forgetful-Functor): *`schema_pin` $\wedge$ `canonicalisation_pin` $\wedge$ `drr_rule_set_pin` unchanged.*

Migration story. CDM v6 to v7: old transactions retain `cdm_version`: v6; new transactions use `cdm_version`: v7; the mapping layer carries a $(v6, v7)$ bidirectional translator with a round-trip test in CI. **Bootstrap.** `L7Pc@genesis` sits in the genesis hash anchor; subsequent updates are L_{13} transactions referencing the prior L_7^{Pc} by `tx_id`.

6 Engineering Principles and Anti-Over-Engineering Vetoes

6.1 The Ten Principles

Principle 6.1 (P_1 Pure-functional ingest). *All ingress functions are total, deterministic, and produce values rather than side effects.*

Principle 6.2 (P_2 Append-only by default). *Mutation is the exception, not the rule. Mutable leaves carry an explicit ADR.*

Principle 6.3 (P_3 Content-addressed identity). *Identifiers are hashes of canonical bytes wherever feasible; mint-on-arrival keys are explicit and persisted in an out-of-workflow dedup table.*

Principle 6.4 (P_4 Bitemporal where the world restates). *Single-axis “as-of” is a defect when the underlying authority can correct itself.*

Principle 6.5 (P_5 Errors are values). *A closed-sum error algebra; no thrown exceptions across module boundaries.*

Principle 6.6 (P_6 Snapshot pinning at every impure boundary). *Reads of mutable state are pinned to a snapshot identifier; never current state.*

Principle 6.7 (P_7 Make illegal states unrepresentable). *Smart constructors and witness types eliminate whole categories of bug at compile time.*

Principle 6.8 (P_8 One canonical record). *L_{13} is the canonical record; all other ledger views are derived projections.*

Principle 6.9 (P_9 Closed enumerations everywhere a free string is tempting). *Free strings are reserved for opaque vendor identifiers and never participate in invariants.*

Principle 6.10 (P_{10} Polymorphism by sum type, not inheritance). *Subtype hierarchies are forbidden in normative types.*

6.2 The Fourteen Vetoes

Each veto V_n has a *falsifying predicate* (the condition under which CI fails) and a *named CI mechanism*. The principle is mechanical: no enforcement is no veto.

#	Veto	Falsifying predicate	CI mechanism
V_1	No three-tier Unit Store as 3 storage entities	≥ 3 ORM tables for “Unit Store”	<code>arch_test::test_unit_store_singl</code>
V_2	Listed-instrument detail not top-level	Top-level taxonomy contains <code>ListedInstrumentDetail</code>	<code>count_check::test_no_listed_top_</code>
V_3	No universal symbology service	Per-vendor identifier mapping declared as a service	<code>arch_test::test_no_symbology_ser</code>
V_4	No per-vendor typed schemas in market storage	Storage layer has vendor-discriminated columns	<code>schema_linter::test_canonical_st</code>
V_5	No per-model typed Greek schemas	<code>Greeks</code> declared per-model rather than via GADT	<code>schema_linter::test_polymorphic_</code>
V_6	No Pricing-DAG topology stored as versioned entity	DAG topology has its own table	<code>schema_linter::test_no_dag_table</code>
V_7	No leaf-count inflation	Top-level taxonomy > 20 leaves	<code>count_check::test_leaf_count_le</code>

#	Veto	Falsifying predicate	CI mechanism
V_8	CDM enum closure is a library version pin	CDM enums stored as a queryable table	<code>schema_linter::test_no_cdm_enum</code>
V_9	Policy is thin sidecar	L_7^P partition exceeds 30 fields	<code>count_check::test_17p_field_cap</code>
V_{10}	SSI lives at boundary, Ledger consumes	Ledger has SSI write API	<code>arch_test::test_no_ssi_write</code>
V_{11}	OrchestrationState is replay-substrate, not data	Economic invariants reference workflow state	<code>arch_test::test_no_econ_orch_dep</code>
V_{12}	No free-text <code>metadata/attributes/extensions</code>	Schema contains <code>Map<String, String></code> field	<code>schema_linter::test_no_free_text</code>
V_{13}	No Trade/Position/PnL/Risk/Account table	Tables of these names exist outside test fixtures	<code>schema_linter::test_no_proj_table</code> (ADR-1 carve-out for L_5 , L_6)
V_{14}	Obligation uniform; regulator is a tag	Regulator-discriminated obligation tables exist	<code>arch_test::test_obligation_unif</code>

Table 2: Anti-over-engineering vetoes with truth conditions and CI mechanisms.

ADR-1 (V_{13} override for L_5 , L_6). L_5 and L_6 are stored caches with single-writer-per-field invariants per StatesHome C_{11} . The cache exists for query performance; the canonical record is L_{13} . ADR-1 documents this carve-out and binds the cache to the cache-invalidation discipline CIv-1, CIv-2, CIv-3 of Theorem 9.5.

7 Cross-Cutting Consistency Laws

The fifteen laws $\Lambda_1, \dots, \Lambda_{15}$ are the cross-cutting properties that must hold across leaf boundaries. Each law states a *strengthened oracle* (a decidable property that the implementation can be tested against) and a *witness class*: *Witnessed* (a direct test exists), *Witnessed via composition* (witnessed under bounded parameters), or *Genuinely unwitnessed* (architectural risk with named owner).

Proposition 7.1 (Λ_1 Lineage Closure). *For every transaction $\tau \in L_{13}$ and every input observation $o \in \tau.\text{lineage}$, o resolves to an L_9 or L_{10} record whose attester is registered in the trust registry, whose attester status is ACTIVE, whose snapshot is retained for the lifetime of τ , and whose aggregation outcome is either `multi_source_consensus` or carries an explicit `single_source_authority_assumption_ref`. Witness class: genuinely unwitnessed under vendor opacity; the surrogate property P_{Λ_1} -VENDOR-OPACITY of Section 13.2 is decidable; the residual is an architectural risk owned by CRO + Head of Reference Data.*

Proposition 7.2 (Λ_2 Snapshot Determinism Closure). *The snapshot used by any deterministic activity is content-addressed: the activity output is a function of the snapshot id alone. Witness class: witnessed (replay test).*

Proposition 7.3 (Λ_3 Settlement-Move Closure). *Every settlement instruction produced from L_{11} maps to exactly one move-stream segment in L_{13} . Witness class: witnessed.*

Proposition 7.4 (Λ_4 Bitemporal Coherence). *For every leaf $L \in C_1 \cup C_4$, the restatement chain length is bounded by the constant κ_{restate} pinned in L_7^{Pb} ; the chain is acyclic and the head is unique. Witness class: witnessed via composition (induction on the bounded chain).*

Proposition 7.5 (Λ_5 Per-Event-Class Conservation). *For every conservative event class $c \in \text{ConservativeClasses}$, the per-class structural sum $\sum_{m \in \tau} m.\text{quantity} \mathbb{K}[m.\text{class} = c] = 0$ for every transaction $\tau \in L_{13}$. Witness class: witnessed.*

Proposition 7.6 (Λ_6 Mandate-as-Unit Conservation). *For every mandate unit, the structurally-typed `FeeReserve` sub-account closes the conservation gap arising from fee crystallisation; high-water marks do not collapse across distinct mandates. Witness class: witnessed.*

Proposition 7.7 (Λ_7 Per-CCP Conservation Scope). *Conservation holds per central counterparty scope; multi-CCP novation is handled by Λ_{15} . Witness class: witnessed.*

Proposition 7.8 (Λ_8 Replay Determinism). *Given identical workflow input and an identical L_{19} snapshot, replay produces bit-identical L_{13} entries. Witness class: witnessed via composition; the cosmic-ray ε bound is provided by erasure-coding (n, k, ε) in C-A₁₂.*

Proposition 7.9 (Λ_9 Forgetful-Functor Composition). *The CDM-to-Ledger mapping factors as a forgetful functor; the dependence relation among leaves is a lattice; the projection μ_{P_3} of Theorem 9.7 is well-defined. Witness class: witnessed.*

Proposition 7.10 (Λ_{10} Workflow-History Replay Coherence). *The transaction id $\text{tx_id} = \text{hash_jcs}(\text{business_event_id}, \text{attempt_seq})$ is independent of run_id ; replay across `ContinueAsNew` boundaries dedupes correctly. Witness class: witnessed.*

Proposition 7.11 (Λ_{11} Calibration / Valuation Model Consistency). *The metamorphic catalogue (vega convexity, forward-rate consistency, AD-versus-bump Greeks, local-vol round-trip) holds across calibration outputs. Witness class: witnessed.*

Proposition 7.12 (Λ_{12} No-Arbitrage Admissibility Closure). *Every certified L_{12} posterior lies in the model-versioned admissible set $\Theta_{\text{AF}}(\text{model_id}, \text{model_version_at_cert_time})$. Witness class: witnessed; the sampler is a Hamiltonian Monte Carlo walk over Θ_{AF} .*

Proposition 7.13 (Λ_{13} Obligation Liveness Closure). *For every obligation with deadline $t_d \leq T_{\max}$, exactly one of DISCHARGED, COMPENSATED, DEFAULTED holds at $t_d + \delta_h$. Witness class: witnessed via composition (TLA^+ Büchi automaton + bounded-horizon simulation + production observability).*

Proposition 7.14 (Λ_{14} Capability-Scope Closure). *Every write attempt against a leaf is gated by a runtime-checked phantom witness whose tag matches a single **writer_cap**; cross-capability writes are rejected. Witness class: witnessed.*

Proposition 7.15 (Λ_{15} Novation Bridge Conservation). *Novations across CCP boundaries decompose as either (i) a union-scope conservation event over the union of the two CCP perimeters, or (ii) two transactions linked by a bridge **Obligation** of kind **NovationBridge**. Witness class: witnessed.*

Final classification. Of the fifteen laws, eleven are witnessed directly, three (Λ_4 , Λ_8 , Λ_{13}) are witnessed via composition under bounded parameters specified in the realism budget, and one (Λ_1) is genuinely unwitnessed under vendor opacity, accepted as architectural risk with the named owner.

8 Determinism Boundary Catalogue

The seventeen boundaries $B_1, \dots, B_{17}$ are the points at which the Ledger’s deterministic kernel reads from a non-deterministic environment. At each boundary the data layer pins the read so that replay reproduces the same value.

#	Boundary	Strategy	CI test ID
B_1	Wall-clock reads (<code>now()</code>)	Captured in workflow start; t_{known} from L_{19}	<code>boundary_ci::test_now_captured_i</code>
B_2	Random / entropy reads	Pinned PRNG seed in workflow input	<code>boundary_ci::test_prng_seed_pinn</code>
B_3	External price / FX / vol feeds	Snapshot-pinned via L_{19}	<code>boundary_ci::test_snapshot_pinne</code>
B_4	External event oracles	Signal-driven; snapshot-pinned	<code>boundary_ci::test_signal_attesta</code>
B_5	Reference data	Bitemporal <code>as_of</code> at workflow start	<code>boundary_ci::test_bitemporal_que</code>
B_6	Settlement infrastructure	<code>SsiSnapshotRef</code> versioning	<code>boundary_ci::test_ssi_snapshot_r</code>
B_7	Calibration filter state	<code>KalmanContinuePayload</code> v1 pinned	<code>boundary_ci::test_kalman_continu</code>
B_8	Workflow scheduling	Temporal-managed deterministic replay	<code>boundary_ci::test_temporal_manag</code>
B_9	CDM enum universe	<code>cdm_version</code> pin	<code>boundary_ci::test_cdm_version_pi</code>
B_{10}	Hash algorithm / canonicalisation	RFC 8785 JCS pinned	<code>boundary_ci::test_jcs_pinned</code>
B_{11}	Operator / human interaction	Append-only verification keys; HSM rotation	<code>boundary_ci::test_keys_append_on</code>
B_{12}	Network / message re-ordering	Idempotency key per signal; OOO handling	<code>boundary_ci::test_signal_idempot</code>
B_{13}	Floating-point determinism	Pinned BLAS / threads / IEEE 754 mode	<code>boundary_ci::test_blas_thread_pi</code>
B_{14}	Storage iteration order	Sorted iteration; no implicit ordering	<code>boundary_ci::test_sorted_iterati</code>
B_{15}	Intra-handler concurrency	Single-threaded handler execution	<code>boundary_ci::test_single_threaded</code>
B_{16}	Unicode / locale	NFC normalisation; UTF-8; fixed locale	<code>boundary_ci::test_nfc_utf8</code>
B_{17}	Test-environment seed	Deterministic seed; CI-pinned	<code>boundary_ci::test_ci_seed_pinned</code>

Table 3: Determinism boundaries with strategy and per-boundary CI test.

9 Compositional Theorems

This section states and proves the six compositional theorems that bind the leaves, the laws, and the boundaries together. The dependency directed acyclic graph $\text{DAG}(\text{A-1}, \dots, \text{A-9}, \text{Lm-1}, \dots, \text{Lm-5}, \text{Th-1}, \dots, \text{T})$ is acyclic; no theorem is its own hypothesis.

9.1 Theorem 1 — Conservation Lifting

Theorem 9.1 (Conservation Lifting). *Let the data layer satisfy:*

1. **D-PARTITION**: *EventClass is the closed sum $\text{EventClass} = \text{ISSUANCE} \uplus \text{ConservativeClasses}$.*
2. **D-CONS-CC**: *For every $c \in \text{ConservativeClasses}$, the per-class structural sum is zero by handler type-tagging (cf. Λ_5).*
3. **D-ISS**: *The issuance handler emits a matched issuer-credit / issuance-debit pair preserving the issuance invariant.*
4. **E-ATOM**: *The executor commits a StateDelta atomically.*
5. **D-IDX**: *The fold $L_{13} \rightarrow (L_5, L_6)$ is a deterministic index.*
6. **D-INIT**: *The genesis state is ledger-balanced.*

Then for every time t and every unit u , the aggregate position over wallets equals the issuance baseline $\pm$ the deterministic balanced moves through t .

Proof of Theorem 9.1. By induction on the L_{13} hash chain. The induction is concrete (over transactions), not abstract (over the sum).

Base case. At genesis, the chain is empty; D-INIT gives the ledger-balanced state, which is the identity for the aggregate.

Step case. Suppose the property holds for the prefix of L_{13} of length $n - 1$. Consider the n -th transaction τ . By D-PARTITION, $\tau.\text{class} \in \text{ISSUANCE} \uplus \text{ConservativeClasses}$:

- If $\tau.\text{class} \in \text{ConservativeClasses}$, then by D-CONS-CC the per-class structural sum on τ is zero, the aggregate is unchanged, and the inductive hypothesis carries.
- If $\tau.\text{class} = \text{ISSUANCE}$, then by D-ISS the matched pair preserves the issuance invariant, the aggregate moves by exactly the deterministic baseline term, and the inductive hypothesis carries.

By E-ATOM the StateDelta is atomic, so no partial state is observable; by D-IDX the fold is deterministic, so the aggregate is well-defined. $\square$

9.2 Theorem 2 — Replay Determinism Lifting

Theorem 9.2 (Replay Determinism Lifting). *Let the data layer satisfy:*

1. **L-LEDGER-DET**: *All Ledger handlers are deterministic given their inputs.*
2. **L-WF**: *Workflow code conforms to the determinism linter L_{WF} (boundary B_{14}); the linter is sound and complete (assumption C-A₁₃); the deployment gate L-COVERAGE enforces conformance.*
3. **C-A9**: *The Temporal workflow-history determinism is registered as a joint Ledger $\times$ Temporal property.*
4. **V-V19**: *Canonicalisation is pinned per RFC 8785 JCS (boundary B_{10}).*

Then, given the same workflow input and the same L_{19} snapshot, replay produces bit-identical L_{13} entries.

Proof sketch of Theorem 9.2. Determinism propagates through the pipeline: input pin (workflow input is a value) plus snapshot pin (refdata reads are content-addressed) plus handler determinism (L-LEDGER-DET) plus workflow determinism (L-WF) yields bit-identical state-delta sequences. JCS canonicalisation (V-V19) ensures that hashing of byte representations is invariant under non-semantic permutations.

The conclusion is framed as a joint Ledger $\times$ Temporal property: workflow-history determinism is a *gated* implication discharged by the linter L_{WF} and the deployment gate. The property is not the Ledger's alone; the joint nature is recorded in C-A₉ and C-A₁₃. $\square$

9.3 Theorem 3 — Obligation Liveness Lifting (Bounded Horizon)

Theorem 9.3 (Obligation Liveness Lifting). *Let the data layer satisfy:*

1. **D-OBL**: *Every obligation is registered with (deadline, predicate, compensation).*
2. **D-DEAD**: *deadline $\leq T_{\max}$, the retention horizon.*
3. **D-DISCH-WIT**: *The discharge predicate is evaluated bitemporally.*
4. **D-COMP-TOTAL**: *The compensation handler is total over the closed enumeration $EventClass \times ObligationKind$ (the populated κ -matrix).*
5. **D-TIMER**: *The Temporal timer fires within bounded delay δ_h .*

Then for every obligation o with deadline $t_d \leq T_{\max}$, exactly one of DISCHARGED, COMPENSATED, DEFAULTED holds at time $t_d + \delta_h$.

Proof sketch of Theorem 9.3. By structural induction on the populated κ -matrix $EventClass \times ObligationKind$ (D-COMP-TOTAL). Each cell of the matrix has a constructor determined by the obligation kind; that constructor terminates either by discharge (predicate satisfied within deadline, by D-DISCH-WIT) or by compensation (the timer fires per D-TIMER and the compensation tower of Section 12.2 runs to a terminal state). Mutual exclusion is structural: the FSM transitions PENDING $\rightarrow$ DISCHARGED, PENDING $\rightarrow$ COMPENSATED, PENDING $\rightarrow$ DEFAULTED are pairwise disjoint at the type level. Boundedness by $T_{\max}$ follows from D-DEAD; the unbounded variant is reclassified as realism-budget axiom RB-3 (cf. Λ_{13}). $\square$

9.4 Theorem 4a — Substantiation Definition

Definition 9.4 (Balance-Sheet Projection). *A balance-sheet line item is the projection of (L_5, L_6, L_{13}) by the named projection function π_{BS} . This is a definition, not a theorem; no proof obligation arises.*

9.5 Theorem 4b — Cache Coherence

Theorem 9.5 (Cache Coherence). *Let the data layer satisfy the cache-invalidation discipline:*

1. **CIv-1 (D-CACHE-WRITE)**: *Every L_{13} commit writes the corresponding L_5, L_6 cache rows in the same atomic StateDelta.*
2. **CIv-2 (D-CACHE-READ)**: *Cache reads use snapshot-pinned **as_of** queries; never current state.*
3. **CIv-3 (D-CACHE-REBUILD)**: *Cache rebuild is a pure function of the L_{13} prefix.*

*together with **D-IDX** (deterministic projection) and **E-ATOM** (atomic commit). Then for every time t :*

$$(L_5, L_6)_t = \pi_{BS}(\text{fold}(L_{13}[\leq t], \text{init})).$$

Proof of Theorem 9.5. By induction on the L_{13} prefix. The base case is the genesis fold (D-INIT inherited from Theorem 9.1). For the inductive step: by CIv-1 the cache write and the L_{13} append are co-atomic (E-ATOM), so the cache after the n -th commit equals π_{BS} of the fold after n commits; by CIv-2 readers see the post-commit cache, not a transient state; by CIv-3 a rebuild from L_{13} alone reproduces the cache, so the cache cannot drift undetected across replicas. $\square$

9.6 Theorem 5 — No-Arbitrage Pricing Lifting

Theorem 9.6 (No-Arbitrage Pricing Lifting). *Let the data layer satisfy:*

1. **D-MODEL**: *Every model m is identified by $(\text{model_id}, \text{model_version})$ pinned per the versioning algebra.*

2. **D-CAL'**: The Kalman posterior $(x_{t|t}, P_{t|t})$ carries *model_version_at_cert_time*.
3. **D-ARB**: $\Theta_{AF} : (\text{model_id} \times \text{model_version}) \rightarrow \text{AdmissibleSet}$ is a model-versioned closed type.
4. **E-GATE'**: The innovation gate downgrades stale records to FSM STALE.

Then every certified L_{12} carries an *arbitrage_certificate* witness asserting $\text{posterior} \in \Theta_{AF}(\text{model_id}, \text{model_version})$.

Proof sketch of Theorem 9.6. The witness type discipline of Section 4.12: the posterior is reachable only inside the **Accepted** branch; the certificate is constructed alongside the posterior; consumers pattern-match and obtain the witness by construction. Model-version pinning eliminates the Heston-2018-vs-Heston-2024 constraint-version edge case: Θ_{AF} is parameterised on the version, not on the bare model id. $\square$

9.7 Theorem 6 — Pillar-3 Projection Lifting

Theorem 9.7 (Pillar-3 Projection Lifting). *Let the data layer satisfy:*

1. **D-LEDGER**: L_{13} and L_{14} are the source of truth.
2. **D-CTR**: L_7^{Pc} contains the accounting classification map.
3. **D-DRR**: The DRR rule-set is pinned per the versioning algebra axis *drr_rule_set_pin*.
4. **D-PROJ**: $\mu_{P_3} : (L_{13}, L_{14}, L_7^{\text{Pc}}) \rightarrow \text{DRR-Pillar3 input}$ is a deterministic projection.

Then every L_{17} *RegulatorySubmission* for Pillar 3 is bit-identically reproducible from $(L_{13}[\leq t_{\text{known}}], L_{14}[\leq t_{\text{known}}], L_7^{\text{Pc}}[\leq t_{\text{known}}])$ via μ_{P_3} .

Proof of Theorem 9.7. By D-LEDGER the Pillar 3 input is a pure function of $(L_{13}, L_{14}, L_7^{\text{Pc}})$; by D-PROJ the function μ_{P_3} is deterministic; by D-DRR the rule-set version is pinned (so the function's behaviour is pinned); by D-CTR the classification map is pinned. The bit-identity follows by JCS canonicalisation of the projection output. $\square$

9.8 Dependency Directed Acyclic Graph

The dependencies among nine axioms ($A-\Lambda_1, A-\Lambda_2, A-\Lambda_3, A-\Lambda_5, A-\Lambda_7, A-\Lambda_8, A-\Lambda_{11}, A-\Lambda_{13}, A-\Lambda_{14}$), five lemmas ($Lm-\Lambda_4, Lm-\Lambda_6, Lm-\Lambda_9, Lm-\Lambda_{10}, Lm-\Lambda_{12}$), and seven theorems ($Th-1, Th-2, Th-3, Th-4a, Th-4b, Th-5, Th-6$) form an acyclic directed graph. No theorem is its own hypothesis. Edges: Th-1 depends on $A-\Lambda_5, A-\Lambda_7$; Th-2 depends on $A-\Lambda_2, A-\Lambda_8$; Th-3 depends on $A-\Lambda_{13}$ and $Lm-\Lambda_{10}$; Th-4a is a definition only; Th-4b depends on Th-1, Th-4a, $Lm-\Lambda_4$; Th-5 depends on $A-\Lambda_{11}, Lm-\Lambda_{12}$; Th-6 depends on Th-1, Th-2, Th-4b. The graph is verified by an automated check at CI time.

10 Realism Budget

The realism budget is the audit-trail of the trust frontier: every guarantee that the framework provides without conditions, and every assumption on which a conditional guarantee rests. Each conditional assumption has four mandatory fields: *detection signal*, *compensating action*, *blast radius*, and a *named owner* (named individual or named role, not an anonymous job title).

10.1 Unconditional Guarantees (U_1 – U_8)

These hold by construction; no external assumption is required.

U_1 Append-only mutation

No leaf mutates a prior version in place. Mutable leaves (L_5 , L_6 , L_{18}) carry an explicit ADR.

U_2 Bitemporal indexing

$C_1 \cup C_4$ leaves carry $(t_{\text{obs}}, t_{\text{known}})$.

U_3 Deterministic identity

All identifiers are content-addressed except the explicit mint-on-arrival keys, which are persisted in an out-of-workflow dedup table.

U_4 Hash-chain tamper-evidence

L_{13} entries chain by `prev_hash`.

U_5 Idempotency on payloads bearing an `IdempotencyTokenSidecar`

Inbound signals carry `signal.idempotency_token`; replays dedupe.

U_6 Schema-pinned validation

All payloads are validated against a versioned schema.

U_7 Single-writer-per-field

Every mutable field carries a `writer_cap` phantom.

U_8 Replay determinism for snapshot consumers

Joint with C-A₉ (Temporal workflow-history determinism).

10.2 Conditional Assumptions (C-A₁–C-A₁₃)

#	Assumption	Owner	Detection signal	Compensation	Blast radius
C-A ₁	Cryptographic primitive soundness	Head of cryptography	Primitive advisories; quarterly review	Emergency rotation; quarantine post-advisory data	All attested data potentially forged
C-A ₂	HSM custody discipline	Head of security operations	HSM tamper alarm; key-usage anomaly	N_{12} kill-switch; rotate keys	Attestation compromised

#	Assumption	Owner	Detection signal	Compensation	Blast radius
C-A ₃	Vendor honesty (per attested vendor)	Per-vendor relationship owner	Cross-vendor disagreement; PnL-explain residual	Quarantine vendor; N_8 multi-source fallback	False attestation passes innovation gating
C-A ₄	Settlement-layer SSI freshness	Settlement-operations team lead	Settlement-fail; virtual-wallet contra-balance break	Refresh SSI; CSDR penalty L_{18} record	Misrouted wires; CSDR penalties
C-A ₅	CDM/ISO 20022/F-pML schema stability within version	CDM/ISO interop lead	Round-trip mapping CI failure	Version-pin freeze; <i>schema</i> axis update on VersionPinSidecar	Historical mappings produce different replay outputs
C-A ₆	Calibration model soundness	Model-validation team lead	PnL-explain residual; cross-asset coherence break	FSM QUARANTINED; downgrade L_{14} quality	Certified state silently wrong
C-A ₇	Authority registry currency (GLEIF / SWIFT / ISO)	Identity-and-trust ops lead	Authority revocation; failed-verify rate above baseline	Trust registry update; replay re-verify	Falsely accept revoked / falsely reject valid
C-A ₈	Closed-system boundary integrity	Architecture review board chair	Code-review enforcement; integration tests refusing non-executor writes	Patch boundary; rerun integration tests	Closed-system property collapses
C-A ₉	Workflow-history determinism (joint Ledger $\times$ Temporal)	Temporal lead	Temporal SDK determinism violations on replay	L_{WF} linter fail; deployment gate	PnL-explain reconstruction fails
C-A ₁₀	Retention sufficiency	Records management + compliance lead	Retention-policy-vs-instrument-lifetime crosscheck at registration	Extend retention; archive promotion	Late-life replay or 7-year audit unanswerable
C-A ₁₁	Canonical-serialiser stability	Canonicalisation owner	RFC 8785 JCS test-suite drift; cross-implementation hash mismatch	Pin canonicalisation_version; freeze migration	Cross-implementation replay claim becomes rhetorical

#	Assumption	Owner	Detection signal	Compensation	Blast radius
C-A ₁₂	Cross-replica integrity (cosmic-ray surrogate)	Storage operations lead	Cross-replica disagreement; erasure-coding integrity test failure	Restore from quorum; replicate	Bit-flip undetected; replay produces phantom answers
C-A ₁₃	L_{WF} linter soundness and completeness	Determinism owner	Linters false-negative discovery via property-test on production replay	Patch linter rule-set; freeze deployment	Th-2 silently false; PnL-explain unreliable

Table 4: Conditional realism-budget assumptions with owner, detection, compensation, and blast radius.

Trust-assumption registry artefact. The thirteen assumptions are not paragraphs; they are rows in a versioned, signed registry (schema: (id, owner_lei, owner_email, review_cadence, last_review, kill_switch_doc_ref, status)) reviewed quarterly at minimum.

11 CDM Gap Analysis

The Common Domain Model (FINOS CDM 6.0.0) was re-fetched from github.com/finos/common-domain-model for this document; fifteen Rosetta source files (approximately 9,100 lines) were compared against the leaf catalogue. Each leaf is classified *Direct*, *Partial*, or *Missing*. The total cross-walk yields eleven Direct, twenty-two Partial, twenty-nine Missing entries (was 14/22/26 before the re-fetch).

11.1 Top-Level Strategic Gaps

#	Gap	Severity	Status	PR units
1	TradeState ↔ StatesHome 3-map alignment	Strategic	Architectural; Th-1 / Th-2 / Th-4b share dependency. 96 round-trip cases; surjective-projection-with-named-axes-lost criterion.	3
2	Calibrated Market Data Layer	Strategic	CDM has the container (TradeState.valuationHistory); content (CalibratedYieldCurve, CalibratedVolSurface, KalmanPosterior, SensitivityJacobian, ValuationRecord) is missing.	5
3	Tokenised Collateral & Backing Attestations	Significant	lifecycle_model = SmartContract on L_1 + EligibleCollateralCriteria extension. DigitalAsset REJECTED as host (live CDM assetType = Other excludes tokenised).	3
4	SBL Recall / Locate / Rehypothection	Operational	ISLA-coordinated; RecallInstruction, LocateConfirmation, RehypInstruction added to PrimitiveInstruction (a type with ≥ 11 optional fields, <i>not</i> a choice).	3
5	SBL Manufactured Payment / Tax Treatment	Operational	ISLA-coordinated.	1

Table 5: Top-level CDM gaps with severity, status, and PR-unit count.

11.2 Per-Leaf Cross-Walk

Leaf	CDM type / path	Status	Gap or extension
L_1	NonTransferableProduct	Direct	Qualified via Qualify_* functions

Leaf	CDM type / path	Status	Gap or extension
L_2	Security extends InstrumentBase	Partial	Missing boardLotSize, votingRights, dividendPolicyRef, corporateActionFeedRef
L_2	ListedDerivative	Missing	No clearingHouse field; CCP-as-unit-identity unaddressable; L_1 extension ExchangeContractSpec
L_3	Party	Direct	ISO 17442 partyId confirmed
L_4	BusinessCenters / DayCountFractionEnum	Direct	—
L_5	TradeState.state	Partial	Mutable status map is Ledger-native shape
L_6	Position	Partial	Six-coordinate decomposition is Ledger-native; SBL coordinates Missing
L_7^P	—	Missing	Ledger-internal (correct)
L_8	LegalAgreement / LegalAgreementTypeEnum	Direct	—
L_9	Observation	Partial	CDM observations scoped to lifecycle, not generic streaming ingest
L_{10}	BusinessEvent / Reset	Partial	Downgraded from Direct: no snapshot-id binding, no executor signature, no hash-chain pointer; SBL events Missing
L_{11}	ISO 20022 / FpML mappings	Partial	Inbound mappings for sese.025, sese.023, camt.053/054
L_{12}	TradeState.valuationHistory (container)	Missing	Content (CalibratedYieldCurve, etc.) Missing
L_{13}	BusinessEvent + TradeState	Partial	Snapshot-id binding, executor signature, hash-chain pointer Ledger-native
L_{14}	Valuation (in TradeState.valuationHistory)	Partial	IPV/AVA fields Ledger-native; container present
L_{15}	—	Missing	Ledger-internal obligation FSM
L_{16}	various per-table	Mixed	Sanctions Missing; Withholding Missing; Corp-actions Schedule Partial; Index Composition Partial; CCP Margin Missing
L_{17}	DRR Pillar 3 / MiFIR / EMIR Refit / SFTR / CFTC Part 49	Partial	Rule-set version pinned per the <i>rule-set</i> axis of the VersionPinSidecar
L_{18}	—	Missing	Ledger-internal break register
L_{19}	—	Missing	Ledger-internal time-source authority

Table 6: Per-leaf CDM cross-walk against FINOS CDM 6.0.0.

Resolved suspect path/type claims. MarginCallInstruction does not exist (correct types: MarginCallBase, Issuance, Response, Exposure in event-common-type.rosetta). IndexTransitionInstruction is in PrimitiveInstruction (a type, not a choice). The enum is LegalAgreementTypeEnum. CDM 6.x uses flat dot-namespaced filenames in rosetta-source/src/main/rosetta/.

Tokenised collateral framing (ADR-10). The combined approach is: `lifecycle_model = SmartContract` on L_1 (ISDA-aligned) plus `EligibleCollateralCriteria` extension in CDM. `DigitalAsset` is rejected as the host: the live CDM condition `assetType = Other` and the doc-string explicit exclusion preclude tokenised assets. The standalone `cdm-tokenisation-lib` proposal is also withdrawn.

Migration discipline. Each Ledger-internal leaf carries a `cdm_native_pending` flag; a migration record is logged in L_{11} when a CDM extension lands. The total distinct-PR-unit count is approximately fifteen.

11.3 Rosetta DSL Fragment Example

The shape of an extension PR (illustrative): the `Reset` type as it stands in CDM 6.x, with the corrected fields:

```
type Reset:
  [metadata key]
  resetValue Price (1..1)
  resetDate date (1..1)
  rateRecordDate date (0..1)
  observations Observation (1..*)
  [metadata reference]
  averagingMethodology AveragingCalculation (0..1)

condition AveragingMethodologyExists:
  if observations count > 1 then averagingMethodology exists
```

Three corrections from earlier drafts: `observations` is `Observation (1..*)` not `ObservationEvent (0..*)`; `averagingMethodology` was missing; the conditional was missing.

12 Operational Floor

The operational floor is the set of structures that turn a specification into a running production system: per-leaf reconciliation pairs, a finite-state machine for breaks, a retention matrix, a service-level matrix, IPV / FRTB AVA fields on the valuation record, and the CSDR penalty schema. The bitemporal restatement orchestration and the saga compensation tower are also in this section because they are operational structures rather than algebraic ones.

12.1 Reconciliation Matrix

Leaf	External authoritative source	Cadence	Tolerance	Break work-flow	Owner
L_1	Counterparty BusinessEvent	CDM	Per trade	Bit-identical	wf-trade-affirm-break Front-office trade support
L_2	Cross-vendor reconciliation	reconcilia- tion	Daily T+0	0 (quarantine)	wf-refdata-break Refdata operations
L_3	GLEIF CDF		Daily	Status match	wf-lei-break Identity-and-trust ops
L_4	Multi-vendor		Daily	0	wf-calendar-break Refdata operations
L_5	Counterparty FpML / sese.025	Per change	Bit-identical	wf-state-break	Lifecycle opera- tions
L_6	CCP statement; custo- dian; triparty	Daily T+1	Per regime	wf-position-break	Middle-office rec- onciliation
L_9	Multi-vendor IPV at FVH	Daily/weekly	FRTB AVA	wf-ipv-break	Independent Price Verification
L_{10}	Calculation agent / index admin	Per event	Authoritative	wf-oracle-break	Lifecycle opera- tions
L_{11}	Inbound ISO 20022	Per mes- sage	Validation	wf-confirm-break	Settlement opera- tions
L_{12}	Cross-engine IPV; PnL- explain	Per cycle	L_7^{Pb} tolerance	wf-cal-break	Quant validation
L_{13}	Hash-chain self-verify; cross-replica	Continuous	0	wf-chain-break	Architecture re- view
L_{14}	Cross-engine IPV; Level 1/2/3 reconcile	Daily	FRTB PLA	wf-valuation-break	Risk
L_{15}	CSA-call; AcadiaSoft; triResolve; ack feeds	Per event	Per kind	wf-obligation-break	Collateral opera- tions
L_{17}	Trade Repository ack	Per sub- mission	Ack match	wf-regsub-break	Regulatory opera- tions
L_{18}	Self-reconciling (records breaks of other leaves)	Continuous	FSM aging thresholds	wf-break-aging	Operations man- agement
L_{19}	Multi-source clock comparison (NTP/PT- P/GNSS)	Continuous	1 ms drift	wf-clock-break	Site reliability

Table 7: Per-leaf reconciliation pair.

12.2 BreakRegister Finite-State Machine

The L_{18} FSM is:

OPEN $\rightarrow$ INVESTIGATING $\rightarrow$ ASSIGNED $\rightarrow$ AGED-1 $\rightarrow$ AGED-3 $\rightarrow$ AGED-5 $\rightarrow$ ESCALATED $\rightarrow$ AT-RISK $\rightarrow$ MAT

Aging thresholds (T+1, T+3, T+5) trigger automatic escalation. The AT-RISK state is entered at $\geq$ EUR 1M exposure (IFRS 13 materiality). The MATERIAL state is entered at $\geq$ EUR 10M and triggers governance committee review. Mandatory four-eyes approval is required on the transition to CLOSED-WAIVED.

12.3 Saga Compensation Tower

A compensation is *trivial* (single-step idempotent state mutation) or *non-trivial* (multi-step, external calls, partial-completion reasoning, distinct discharge contract). Non-trivial compensations are child workflows; the ruling (ADR-8) is structural.

The three-tier tower per non-trivial obligation:

Tier 1 — Compensation workflow

Started as a child of `ObligationWorkflow` on timer-fire. Own deadline (original_deadline+compensation_horizon), own retry policy, own discharge contract.

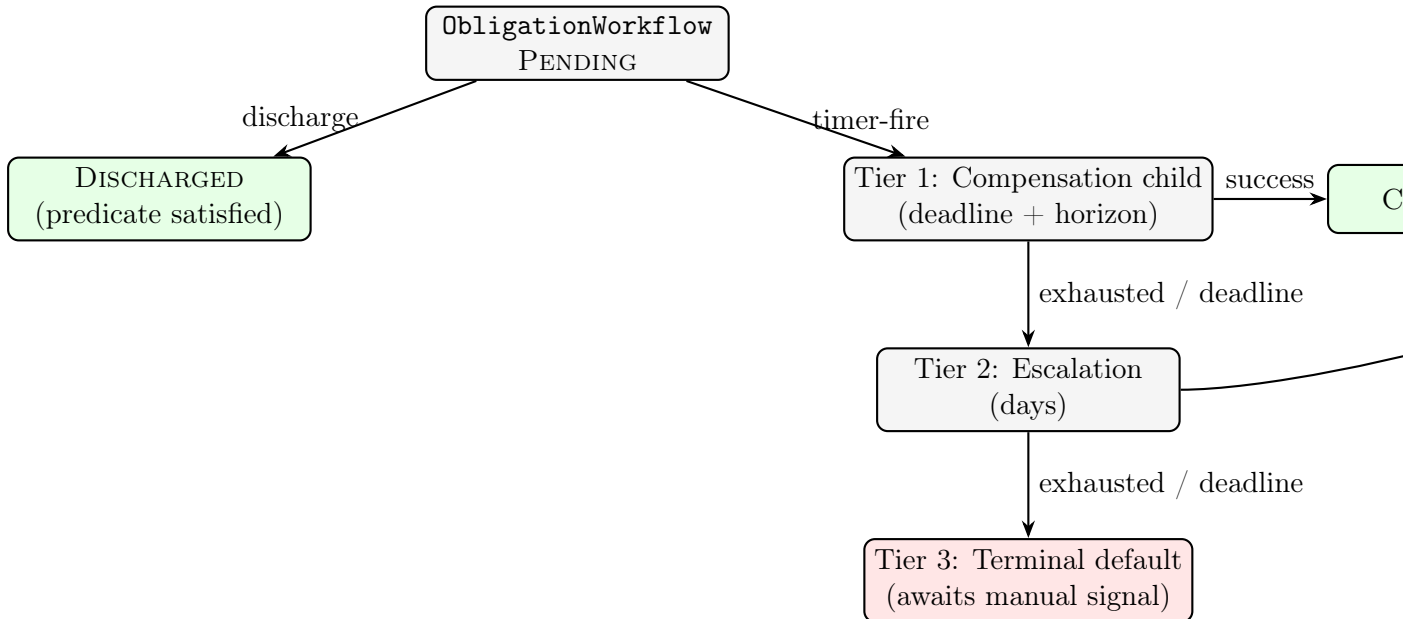
Tier 2 — Escalation workflow

Started on Tier 1 exhaustion. Own deadline (typically days). Examples: buy-in escalation to manual-ops queue plus regulatory notification; manufactured-payment under-withhold-then-true-up failure to tax-authority correspondence.

Tier 3 — Terminal default

Started on Tier 2 exhaustion. Records the obligation as `TERMINALDEFAULT`, fires regulatory disclosures, and waits indefinitely on a manual-ops signal.

TikZ diagram of the tower:



Late-discharge race policy. The race — a discharge signal arriving *after* the deadline timer fired and compensation began — is resolved per obligation kind. SBL recall, manufactured payment, locate confirmation, and buy-in confirmation: *Reject* (deadline sacrosanct). Settlement and margin call: *Cancel-compensation if compensation has not externalised; otherwise queue-and-reconcile*. Regulatory submission ack: *Queue-and-reconcile*. Coupon / income event: *Cancel-compensation*. The policy table is pinned per kind via the *schema* axis of the Version-PinSidecar.

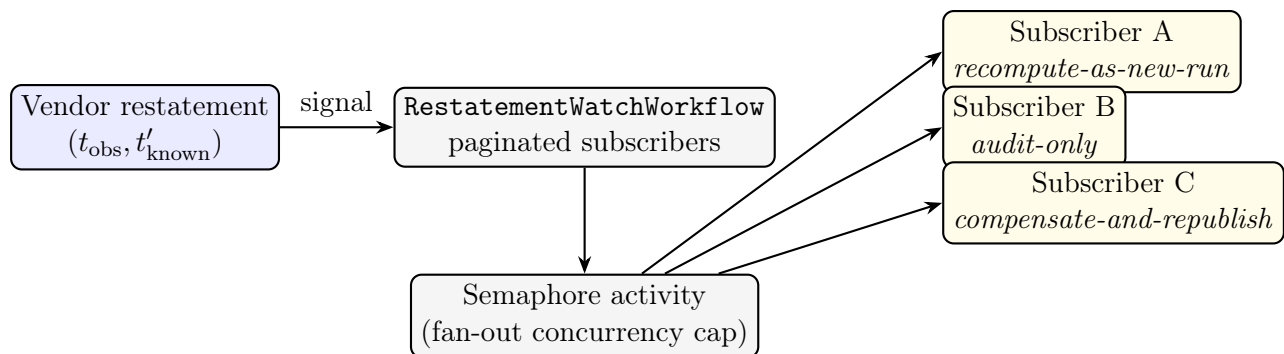
12.4 Bitemporal Restatement Orchestration

A single `RestatementWatchWorkflow` runs per `(observable_class, vendor)` tuple, where the observable class lies in the closed sum

`{raw_market, calibrated, fixing, corp_action, calendar, party_lei, ssi, tax_form}`.

The workflow id format is `restate-watch:{observable_class}:{vendor}`. Each instance subscribes to “new- t_{known} -for-prior- t_{obs} ” events from the snapshot store, holds a paginated subscriber list in workflow state, fans out signals, tracks acknowledgements, retries unacked subscribers per a backoff policy, and continues-as-new at 8000 events or 30 days.

TikZ diagram of the workflow:



A subscriber registers at the moment of consumption with a typed predicate `{observable_id, t_obs_range}`. On each restatement matching the predicate, the subscriber takes one of three stances: *audit-only*, *recompute-as-new-run*, or *compensate-and-republish*. The workflow does not decide; it routes. Calendar restatements use this same workflow under `observable_class = calendar`; the mode-1 pin guarantees that already-fired timers are untouched.

12.5 Retention Matrix

Leaf	SOX	MiFIREMIR	CFTC 49	BCBS	FRTB	CASS	DORA RTO/RPO	GDPR conflict rule
L_1	7y	5y	5y	life+5y	cycle	cap-hist	—	4h / 0 LEI retained
L_2	7y	5y	5y	life+5y	—	—	—	4h / 1h —
L_3	7y	5y	5y	life+5y	—	—	—	4h / 1h LEI retained; erasure de- ferred

Leaf	SOX	MiFIREMIR	CFTC 49	BCBS	FRTB	CASS	DORA RTO/RPO	GDPR con- flict rule	
L_4	7y	5y	5y	life+5y	—	cap- hist	—	24h / 24h	—
L_5	7y	5y	5y	life+5y	cycle	cap- hist	5y	1h / 0	—
L_6	7y	5y	5y	life+5y	cycle	cap- hist	5y	1h / 0	—
L_7^P	7y	5y	5y	life+5y	cycle	cap- hist	—	24h / 24h	—
L_8	7y	5y	5y	life+5y	—	—	—	24h / 24h	Counterparty info retained
L_9	7y	5y	5y	life+5y	—	cap- hist	—	4h / 1h	—
L_{10}	7y	5y	5y	life+5y	cycle	—	—	1h / 0	—
L_{11}	7y	5y	5y	life+5y	cycle	—	—	1h / 0	—
L_{12}	—	5y	—	—	—	cap- hist	—	4h / 1h	—
L_{13}	7y	5y	5y	life+5y	cycle	cap- hist	5y	0 / 0	Pseudonymisation
L_{14}	7y	5y	—	—	cycle	cap- hist	—	1h / 0	—
L_{15}	7y	5y	5y	life+5y	cycle	—	5y	1h / 0	—
L_{16}	7y	5y	5y	life+5y	cycle	—	—	24h / 24h	—
L_{17}	—	5y	5y	life+5y	—	cap- hist	—	4h / 1h	—
L_{18}	7y	5y	5y	—	cycle	—	5y	24h / 24h	—
L_{19}	7y	5y	5y	life+5y	—	—	—	0 / 0	—

Table 8: Per-leaf retention horizons (SOX 7y; MiFIR 5y; EMIR 5y after expiry; CFTC Part 49 life of swap + 5y; BCBS 239 through-the-cycle; FRTB capital history; CASS / Rule 15c3-3 5y).

Perpetual-issuance edge case. For instruments with no maturity (perpetual bonds, ever-green mandates), the retention horizon is $\max(\text{applicable_regulation_horizon}, \text{position_state_non_zero} + 7y)$.

GDPR-minimisation conflict resolution rule. Regulatory retention preempts the GDPR right-to-erasure for fields covered by a retention obligation; non-mandatory personal data is minimised at issue.

12.6 Service-Level Matrix

Leaf	p50	p99	Degraded mode	DORA RTO	DORA RPO
L_1	100 ms	1 s	Reject; manual review	4 h	0
L_2	1 s	10 s	Stale-allowed marker; FSM STALE on dependents	4 h	1 h
L_3	1 s	10 s	Stale-allowed for non-critical; reject new trades on lapsed LEI	4 h	1 h
L_4	1 s	10 s	Reject pricing; FSM STALE	24 h	24 h
L_5	10 ms	100 ms	Reject reads; quorum-based	1 h	0
L_6	10 ms	100 ms	Reject reads; quorum-based	1 h	0
L_7^P	100 ms	1 s	Last-known-good with quality flag	24 h	24 h
L_8	100 ms	1 s	Reject new trades; existing unaffected	24 h	24 h
L_9	1 ms	10 ms	FSM STALE; quality flag	4 h	1 h
L_{10}	100 ms	1 s	Defer to bilateral fallback; manual reconciliation	1 h	0
L_{11}	100 ms	1 s	Stale-allowed; reconciliation queue	1 h	0
L_{12}	1 s	10 s	FSM STALE; downgrade L_{14} quality	4 h	1 h
L_{13}	10 ms	100 ms	Reject commit; defer to manual review	0	0
L_{14}	100 ms	1 s	FSM STALE	1 h	0
L_{15}	100 ms	1 s	Saga compensation; escalation	1 h	0
L_{16}	1 s	10 s	Stale-allowed	24 h	24 h
L_{17}	1 s	10 s	Queue; submit on recovery	4 h	1 h
L_{18}	100 ms	1 s	Aging proceeds; manual escalation	24 h	24 h
L_{19}	100 us	1 ms	Refuse pricing; FSM STALE	0	0

Table 9: Per-leaf p50 / p99 ingress SLA, degraded mode, DORA RTO/RPO.

T+1 affirmation. L_{11} affirmation must complete by 9pm ET on T+0. SLA p99 cutoff: 9pm ET on T+0 minus the settlement-orchestration buffer.

T+0 settlement (US equities post-2024). L_{11} confirmation must arrive by 11:30am ET on T (DTCC cutoff). Late confirmation triggers **wf-confirm-break** with a CSDR penalty obligation if cross-border.

12.7 IPV / FRTB AVA on L_{14}

The valuation record carries the IPV / FRTB AVA fields enumerated in Section 4.14: **fair_value_level**, **ipv_status**, **ipv_variance**, **ipv_source_id**, **prudent_valuation_adjustment_components** (with the seven sub-components: market-price uncertainty, close-out cost, model risk, concentrated position, future administrative costs, early termination, operational risk), **unobservable_inputs[]**, **unobservable_input_sensitivity[]**.

12.8 CSDR Penalty Schema

The CSDR penalty kind is encoded as `obligation_kind = CSDR_PENALTY` with the schema `(rate_basis_points, days, source_lei, currency)`. The penalty fires on settlement-fail attestations from custodians or CCPs.

12.9 Lineage Cursor

The lineage cursor is a typed graph projection over $L_{13} \oplus L_{12} \oplus L_9 \oplus L_{10} \oplus \text{envelopes} \oplus \text{VersionPinSidecar} \oplus \text{capabilities}$ with materialised forward and reverse edges, supporting SOX §404, BCBS 239 §3, DORA Article 8, and IFRS 13 Level 3 query paths. Implemented as Datalog over content-addressed identities.

13 Property-Based Testing Program

The data layer is verified by a stratified test pyramid in which every layer has a target count, a property kind, a generator strategy, and a coverage gate. Without a stratified pyramid the law Λ_n becomes a paragraph rather than a property; without explicit coverage gates the test corpus drifts.

13.1 Test Pyramid Declaration

Layer	Target count	Property kind	Generator strategy	Coverage gate
Unit (Φ_n per leaf)	~ 640 (16 families $\times \sim 40$)	Type-level	Hypothesis proptest over closed sums	100% (mandatory)
Property (per Λ_n)	15	Cross-layer	Stratified per coverage targets	Per-stratum assertions
Mutation	~ 750 (15 ops $\times 50$ sites)	Mutation testing	<code>mutmut</code> / <code>cosmic-ray</code>	Stratified kill-rate
Saga / chaos	~ 30	End-to-end	<code>temporal-test-server</code> + Buggify	Determinism replay
Boundary (B_1 – B_{17})	17	I/O integration	Recorded fixtures	Per-boundary CI test
Regression fixture corpus	≥ 40	Characterisation	Recorded-replay	100% retained as guards

The CI gate fails on any layer below its target count.

13.2 Property Oracle for Λ_1

The genuinely-unwitnessed status of Λ_1 stands on the impossibility of provably establishing vendor honesty under composition. The runtime obligation is decidable, however, via the trust-registry surrogate:

```

property P-Lambda1-VENDOR-OPACITY (tx: Transaction):
  for each input observation o in tx.lineage:
    assert TrustRegistry.contains(o.attestor)
    assert TrustRegistry[o.attestor].status == ACTIVE
    if o.aggregation_outcome == single_source_authority:
      assert o.single_source_authority_assumption_ref
        in TrustRegistry
      assert TrustRegistry.last_review[o.attestor]
        within review_cadence
    else:
      assert o.aggregation_outcome
        in {multi_source_consensus, quarantined}

```

The property runs at every commit boundary, in $\mathcal{O}(|\text{tx.lineage}| \cdot \log |\text{TrustRegistry}|)$. False under: unregistered attestor, killed attestor, lapsed review, single-source admission without registered exception. Kill-switch: a registry status flip from ACTIVE to KILLED quarantines all post-attestor data and triggers replay re-verify against alternate sources.

13.3 HWM-Cross-Mandate-Collapse Fixture

```

fixture HWM-COLLAPSE-1 (Lambda_6 violation candidate):
    setup:
        mandate_A = create_mandate(qis=A, hwm=100, fee_pct=0.20)
        mandate_B = create_mandate(qis=B, hwm=100, fee_pct=0.20)
        move(client_C -> mandate_A, units=10)
        move(client_C -> mandate_B, units=10)
    trigger:
        nav_update(mandate_A, new_nav=120)
        nav_update(mandate_B, new_nav=80)
        crystallise_fee(mandate_A)
        nav_update(mandate_A, new_nav=80)
    assertion:
        assert hwm[mandate_A] != hwm[mandate_B]
        assert fee_accrual[mandate_A] >= 4
        assert fee_accrual[mandate_B] == 0

```

Plus four variant fixtures: cross-currency QIS; manufactured-payment cross-jurisdiction; tax-treatment-divergent QIS; mandate-as-unit-being-superseded mid-crystallisation.

13.4 Mutation Kill-Rate Floors

Stratified kill-rate floors by (Λ -cluster $\times$ mutation operator). Bold cells are critical: survivors fail the build. Other cells are advisory floors; survivors generate review-required findings.

Cluster	CONSBND	VAC	CLK	CCH	NPJ	CAN	CDM	CAP	LAT	FCT	AGG	DFL	BIA	SHR	
$\Lambda_5/\Lambda_6/\Lambda_7/\Lambda_{15}$	95	80	90	50	90	95	80	80	80	50	80	95	90	70	60
$\Lambda_1/\Lambda_2/\Lambda_8$	80	80	80	80	95	80	80	80	80	70	80	80	70	70	70
$\Lambda_{11}/\Lambda_{12}$	80	80	80	70	80	80	80	70	70	95	80	95	90	90	70
Λ_{13}	80	80	80	80	80	80	80	80	95	80	80	80	70	90	90
Λ_4	80	80	80	80	95	80	80	80	80	80	80	80	70	70	70
Λ_{14}	80	80	80	80	80	80	80	95	80	80	80	80	70	70	70

Table 10: Mutation kill-rate floors (%) by (Λ -cluster $\times$ operator). Bold cells are critical.

14 Goodhart Hardening

A specification is a target; a target invites optimisation; optimisation against a target without honesty is Goodhart’s law in action. The data layer hardens against five named traps. Each trap is paired with a detection mechanism.

#	Trap	Detection / avoidance mechanism
GT ₁	Snapshot stub-swap	Boundary-integrity production test: pull N committed transactions from the production snapshot store; replay; assert byte-identical
GT ₂	Mutation-set exclusion	Fifteen mutation operators with mandatory survivor reporting and stratified kill-rate targets (Section 13.4)
GT ₃	Biased generators	First-class coverage targets per stratum: 5% near-arbitrage-boundary, 10% deadline-near-fire, 10% multi-CCP, 5% manufactured-payment-cross-jurisdiction. CI-asserted.
GT ₄	Aggregation-masking	Meta-property: aggregation tests must pass at every per-class projection, not only globally
GT ₅	Type-system witness laundering	Three-layer defence: (a) type system with strict checker plus AST lint banning <code>cast</code> / <code>__new__</code> outside the constructor module; (b) runtime checker on every write attempt; (c) adversarial Hypothesis test attempting reflection / <code>__post_init__</code> bypass. Closed witness inventory W_1 – W_5 enumerates construction, consumer, and elimination sites

The witness inventory W_1 `arbitrage_certificate`; W_2 `snapshot_certificate`; W_3 `conserving_transaction` / `balanced_transaction`; W_4 `registered_unit_id`; W_5 `calibrated_state` (GADT-keyed by model). Each witness has a single construction site and is consumed by exhaustive pattern match.

15 Surfaced Disagreements: Plain Rulings

The five structural disagreements surfaced during the construction of this document are recorded as plain rulings. The “tension box” format (presenting two positions in parallel without a decision) is forbidden.

*D*₁ **Leaf count**

Resolution: nineteen leaves (Sections 3). ADR-2, ADR-3, ADR-4. No alternative tolerated.

*D*₂ **Tokenised collateral framing**

Resolution: `lifecycle_model = SmartContract` on *L*₁ plus `EligibleCollateralCriteria` extension in CDM. `DigitalAsset` excluded by live CDM. ADR-10. No alternative tolerated.

*D*₃ **Surrogate witnesses**

Resolution: Λ_1 is genuinely unwitnessed under vendor opacity, accepted as architectural risk with named owners (CRO + Head of Reference Data); the runtime obligation is discharged by the property oracle of Section 13.2. $\Lambda_4, \Lambda_8, \Lambda_{13}$ are witnessed via composition with bounded parameters specified in the realism budget.

*D*₄ **Stored caches** *L*₅, *L*₆

Resolution: ADR-1 carve-out of veto *V*₁₃ citing StatesHome *C*₁₁. Cache-invalidation discipline CIv-1, CIv-2, CIv-3 per Theorem 9.5. No alternative tolerated.

*D*₅ **Compensations**

Resolution: three-tier saga tower (compensation child workflow → escalation workflow → terminal default). Trivial compensations remain activities. ADR-8.

16 Architecture Decision Register

ADR	Title	Veto / Leaf	Justification
ADR-1	L_5/L_6 stored cache override of V_{13}	V_{13}	StatesHome C_{11} single-writer invariant; cache-invalidation discipline CIv-1/2/3 per Th-4b
ADR-2	L_{17} RegulatorySubmission elevation	New leaf	CFTC, EMIR Refit, SFTR, SLATE, MiFIR RTS 22, FRTB Pillar 3; DRR rule-set version pin
ADR-3	L_{18} BreakRegister elevation	New leaf	SOX §404; BCBS 239 §3; DORA Article 8
ADR-4	L_{19} ClockAuthority elevation	New leaf	Time-translation invariance carrier; replay determinism
ADR-5	Mode-1 calendar amendment pin	L_4	Preserves time-translation invariance; calendar republication produces new bitemporal record but does not invalidate prior pins
ADR-6	Canonicalisation pin (RFC 8785 JCS / Protobuf canonical / CBOR per RFC 8949)	VersionPinSidecar / B_{10}	Cross-implementation replay determinism
ADR-7	<code>tx_id</code> formula excludes <code>run_id</code>	L_{13}	<code>ContinueAsNew</code> changes <code>run_id</code> ; old formula breaks idempotency
ADR-8	Compensation workflows for non-trivial obligations	L_{15}	Three-tier saga tower; trivial compensations remain activities
ADR-9	Single-source admission via $N_{8.2}$ explicit assumption ref	L_9	Vendor honesty C-A ₃ named assumption; multi-source default
ADR-10	Tokenised collateral via <code>lifecycle_model = SmartContract</code> on L_1 + <code>EligibleCollateralCriteria</code> extension	L_1 / CDM	<code>DigitalAsset</code> rejected by live CDM (<code>assetType = Other</code> excludes tokenised)
ADR-11	Public verification keys append-only	B_{11}	HSM rotation discipline; replay re-verify of old envelopes
ADR-12	$L_7^{\text{Pa/b/c}}$ partition with ≤ 30 fields per partition	L_7^{P}	Veto V_9 enforcement via CI

Table 11: Architecture decision register.

17 Lines-of-Code Budget

A specification that does not bind itself to an implementation budget invites over-engineering. The budget below is normative; an implementation that exceeds the hard cap signals a defect in the specification, not (only) in the implementation.

Component	Budget	Hard cap
Core Ledger types (closed sums + smart constructors + parsers)	3,000 LoC	5,000 LoC
Core Ledger handlers (per (event_class × obligation_kind) cell)	2,500 LoC	4,000 LoC
Core executor + chain logic	1,500 LoC	2,500 LoC
Total core implementation	≤ 7,000 LoC	≤ 10,000 LoC
Vendor adapters (one per vendor / source; ~ 10 sources)	~ 500 LoC each	~ 6,000 LoC
Test suite (unit + property + saga + regression)	~ 12,000 LoC	~ 20,000 LoC
Total system (core + adapters + tests)	≤ 25,000 LoC	≤ 30,000 LoC

Refactoring discipline. Every new abstraction must justify itself against the cap. If the implementation overflows the hard cap, the specification is wrong: review for premature abstraction, leaf-count creep, or a missing closed-sum.

18 Verification Approach

The data layer is verified by ten audit steps. The first six are mechanical (CI-enforced); the last four are governance steps performed at quarterly cadence at minimum.

1. **Per-veto CI mechanism inventory.** For each V_n , verify the falsifying-predicate test exists, runs on every commit, and has been observed to fail at least once during development.
2. **Per-boundary CI test.** For each B_n , verify the boundary integration test runs on every commit and the recorded fixture is byte-identical to the production snapshot.
3. **Theorem hypothesis totality.** For each $\text{Th-}n$, verify every quantifier range is explicit and every base case is stated; the dependency DAG is acyclic.
4. **Surrogate parameters specified.** For each witnessed-via-composition law, verify the parameter is recorded in L_7^{Pb} and observable in production: Λ_4 retention horizon κ_{restate} ; Λ_8 erasure-coding (n, k, ε) ; Λ_{13} T_{max} and δ_{h} .
5. **CDM cross-walk re-fetch.** The Direct / Partial / Missing labels reflect live FINOS CDM 6.0.0; suspect path/type claims have been resolved against the canonical source.
6. **Versioning Algebra composition rules.** For each Λ_n , the required pin axes are stated; mutation tests verify that flipping a pin breaks the law.
7. **Operational floor matrices are normative.** Reconciliation, retention, SLA, IPV: each row is owned by a named team; each cell has been reconciled against an external authoritative source.
8. **Trust-assumption registry artefact.** The thirteen conditional assumptions are rows in the registry; the schema, review cadence, and kill-switch are real (not paragraphs).
9. **Mutation kill-rate stratified report.** Bold cells of Section 13.4 are CI-blocking; advisory cells are reviewed quarterly.
10. **Goodhart trap detection.** Each of $\text{GT}_1, \dots, \text{GT}_5$ has at least one production-attempted test that has failed during development and been remediated; the witness inventory W_1 – W_5 is current.

19 Glossary

The following terms are used throughout this document with specific technical meanings.

Attestor

An external authority (vendor, regulator, calculation agent, custodian, CCP) whose statements enter the Ledger only through a bitemporal observation leaf with a verified signature.

BCBS 239

Basel Committee paper on risk data aggregation and reporting; cited for the through-the-cycle retention requirement and break-management discipline.

Bitemporal axes

The pair $(t_{\text{obs}}, t_{\text{known}})$ on every record in $C_1 \cup C_4$; mandatory two-axis indexing.

BreakRegister

Leaf L_{18} ; mutable break record with the FSM of Section 12.2.

CDM

FINOS Common Domain Model, expressed in the Rosetta DSL.

ContinueAsNew

Temporal SDK construct that closes a workflow run and starts a new run with a transformed payload, preserving idempotency keys.

Conservation

The ledger invariant $\sum_w w(u) = 0$ per unit class; the structural sum is enforced by handler type-tagging.

CRR Article 105 PVA

EU Capital Requirements Regulation Article 105 prudent valuation adjustments; reflected in L_{14} `prudent_valuation_adjustment_components`.

CSDR

EU Central Securities Depositories Regulation; settlement-fail penalty regime on cross-border settlement.

DORA

EU Digital Operational Resilience Act; Article 8 break-management; RTO/RPO matrix in Section 12.6.

DRR

Digital Regulatory Reporting (CDM-native rule sets); pinned per the versioning algebra axis `drr_rule_set_pin`.

EMIR Refit

EU European Market Infrastructure Regulation refit; transaction reporting.

FpML

Financial products Markup Language; XML schema for OTC derivatives; an inbound wire format for L_{11} .

FRTB AVA

Fundamental Review of the Trading Book Additional Valuation Adjustments; the seven AVA categories on L_{14} .

FRTB PLA

Fundamental Review of the Trading Book PnL Attribution test.

IPV

Independent Price Verification.

ISDA Notices Hub

Industry hub for ISDA Master Agreement default and termination notices.

JCS canonicalisation

JSON canonicalisation per RFC 8785; pinned at boundary B_{10} .

KIKO

Knock-In / Knock-Out; an option lifecycle pattern.

 Λ -laws

The fifteen cross-cutting consistency laws of Section 7.

Mandate-as-unit

Modelling pattern in which a mandate (e.g. a managed account or QIS strategy) is itself a unit in $\mathcal{U}$; cf. [1, StatesHome].

MiFIR

EU Markets in Financial Instruments Regulation; transaction reporting under RTS 22.

MoveStream

Leaf L_{13} ; canonical hash-chained sequence of **BalancedTransaction**.

Pricing DAG

Directed acyclic graph of pricing dependencies (cf. [2, §6]).

ProductTerms

Leaf L_1 ; StatesHome map 1.

QIS Quantitative Investment Strategy unit type.

RestatementWatchWorkflow

The bitemporal-restatement orchestration of Section 12.4.

SFTR

EU Securities Financing Transactions Regulation.

SLATE

FINRA Securities Lending and Transparency Engine.

SOX

US Sarbanes-Oxley Act §404; control discipline for financial reporting.

StatesHome 3-map

The three maps **ProductTerms** (L_1), **UnitStatus** (L_5), **PositionState**[(w, u)] (L_6).

Θ_{AF} Model-versioned no-arbitrage admissible parameter set; $\Theta_{\text{AF}} : (\text{model_id} \times \text{model_version}) \rightarrow \text{AdmissibleSet}$.

UnitStatus

Leaf L_5 ; StatesHome map 2.

 Φ -invariants

Per-leaf invariant families with type (T), workflow (W), and composition (C) variants.

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References

- [1] R. Delloye, “The Ledger: A Closed-System Framework for Financial Settlement — Unified Design Specification: Architecture, Lifecycle, and CDM Integration,” v10.3, March 2026.
- [2] R. Delloye, “The Ledger Valuation Stack: Pricing, Greeks, Calibration, PnL Attribution, and Temporal Orchestration,” v1.0, March 2026.